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Survey paper

Computer vision framework for crack detection of civil infrastructure—A review

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ABSTRACT

Civil infrastructure (e.g., buildings, roads, underground tunnels) could lose its expected physical and functional conditions after years of operation. Timely and accurate inspection and assessment of such infrastructures are essential to ensure safety and serviceability, e.g., by preventing unsafe working conditions and hazards. Cracks, which are one of the most common distress, can indicate severe structural integrity issues that threaten the safety of the structure and people in the environment. As such, accurate, fast, and automatic detection of cracks on structure surfaces is a major issue for a variety of civil engineering applications. Due to advances in hardware data acquisition systems, significant progress has been made in the automatic detection and quantification of cracks in recent decades. This paper provides a comprehensive review of the research progress and prospects in computer vision frameworks for crack detection of civil infrastructures from multiple materials, including asphalt, concrete, and metal-like materials. The review encompasses major components of typical frameworks, i.e., data acquisition techniques, publicly available datasets, detection algorithms, and evaluation metrics. In particular, we provide a taxonomy of detection algorithms with a detailed discussion of the advantages, limitations, and application scenarios of the methods in each category, as well as the relationships between methods of different categories. We also discuss unsolved issues and key challenges in crack detection that could drive future research directions.

1. Introduction

Civil infrastructure such as highways, pavement, bridges, tunnels, etc., may lose their expected physical and functional conditions due to long-term operation and exposure to severe scenarios associated with internal defects, aggressive usage, and continuous load changes. Inspection and condition assessment of civil infrastructures are necessary to ensure safety and serviceability (Wu and Liu, 2021; Taheri, 2019; Salehi et al., 2019), where a lack thereof could lead to disastrous incidents. For example, at 18:01 pm on August 1, 2007, the I-35 W Mississippi River Bridge connecting the east side of downtown Minneapolis and Mars-Holmes suddenly collapsed (O'Connell et al., 2007). Dozens of cars fell into the river, causing 13 fatalities and 145 people injured. The accident is one of the most serious bridge collapse incidents in the United States in recent decades caused by non-natural disasters or external forces. It was reported that 68% of the major roads and highways in the USA are in poor condition, which increased vehicle operating costs by USD 61 billion annually. It is estimated that an additional USD 130 billion is required to improve the pavements' condition (Gopalakrishnan, 2018). To prevent unsafe working conditions and hazards, there is a need for continuous and periodic inspection of the infrastructures, which calls for the development of accurate and fast inspection methods.

One of the most common types of defects in civil infrastructure is cracking, which evolves rapidly under the impact of heavy traffic, aging of materials, and drastic environmental changes. Cracks often develop and extend in three-dimension on the surface and inside the structure, which eventually causes local failure or fracture. Since cracking can accelerate the deterioration process, the occurrence and severity of cracking serve as important indicators of the need for maintenance. Crack detection is the first step for several diagnosis and inspection processes. The detection of cracks not only enables the prediction of future conditions and predictive maintenance, but also supports investment planning through the management of limited repair resources, thus ensuring the infrastructure still meets its service requirements with limited maintenance effort.

Conventional methods for inspection of cracks on infrastructure surfaces rely on professional engineers with rich experience to record

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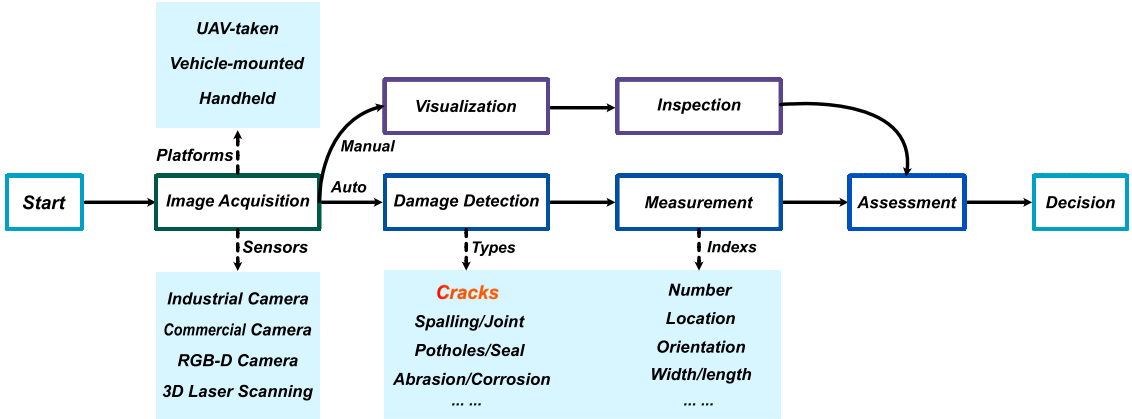

Fig. 1. Computer vision pipeline for manual/automatic inspection of civil infrastructures.

the irregular conditions on structures at regular times or after disasters. They manually assess the physical and functional conditions of infrastructures to ensure their safety and serviceability still meet the expected requirements. Although the manual inspection is widely practiced, it is time-consuming, labor-intensive, and costly to perform manual inspection and evaluation methods at fixed/regular intervals. Also, manual inspection is prone to subjectivity and can potentially hinder the dependability of the quantitative analysis as it completely depends on the specialists' knowledge and experience. With the progress of visual sensing technologies (e.g., digital cameras), which are low cost, high resolution, reliable, and easy-to-use, vision-based methods for crack detection and structural health inspection are gaining popularity. Fig. 1 shows a typical vision-based inspection pipeline that consists of five processes, i.e., 'Acquisition–Detection–Measurement–Assessment–Decision' (ADMAD). ADMAD helps agencies monitor the civil infrastructure, optimize limited maintenance resources, and prevent catastrophic accidents. Visual systems can be permanently installed and are capable of continuously monitoring structural health. To facilitate efficient infrastructure maintenance, it is critical to achieve accurate, reliable, and automatic crack detection, segmentation, and classification. As such, a large number of automatic crack detection algorithms have been developed over the last two decades. In particular, recent years have witnessed remarkable developments made by deep learning techniques for crack detection.

A typical automated crack detection framework takes a video or an image as input and returns one or more regions or pixels as output representing the defect or damage of materials. The type, number, width, and length of the cracks on the structural surface show the earliest degradation level of the structures. However, achieving accurate and efficient detection and quantification of the cracks (e.g., the type, severity, and extent of surface cracking) on civil infrastructure elements (e.g., buildings, bridges, tunnels, pipes, and roads) is challenging. This is because crack detection is affected by many factors such as the complex geometries and typologies of cracks (shape, magnitude), varying degrees of cracking, irregular patterned background, camera distance, environmental conditions (lighting and shading), and limited data, etc. Specifically, (1) The geometries and typologies of cracks (shape, magnitude) are complex and challenging to describe with quantitative features. Various types of cracks show diverse morphology (e.g., line, graph, hole, and other irregular geometries), magnitudes (e.g., number, width, length, etc.), shapes (e.g., longitudinal, transverse, alligator), and backgrounds (e.g., complex scene, similar texture, uneven illumination, etc.). (2) There are various background noises of different patterns and textures. Taking the asphalt pavement images as an example, the pothole, ruts, looseness, and subsidence could appear very similar to real cracks. Also, concrete bridges may contain spalling/joint spall, efflorescence, and holes, while metal-like beams usually contain joints, seals, bearings, and other connections that present more complex geometry. The existence of such noises makes it difficult to distinguish cracks from the backgrounds. (3) The process of capturing crack images also affects the subsequent detection process. Specifically, the data acquisition system needs to limit and adjust the internal and external parameters, including distance, viewing angle, focal length, lighting conditions, etc. These factors will affect the quality of resultant data. (4) The crack detection problems also suffer from limited public datasets with labels. The datasets from different application domains (e.g., asphalt pavement and concrete bridges) have significantly different characteristics. Images of a crack dataset typically have severe label noise, which could be generated by inter-observer variability or errors due to human annotators or computer-generated labels. As such, many existing works allow a tolerance margin in measuring the coincidence between the detected cracks and the ground truth (Shi et al., 2016; Li et al., 2017a; Ai et al., 2018). Also, due to the characteristics of cracks, label imbalance in training data is commonly observed, e.g., the crack pixels are much fewer than the background pixels. Due to label imbalance, the training model converges to major classes (background) faster than minor classes (cracks), such that most examples in the major class exhibit small losses. That is, there is a risk of discarding most examples in the minor class. However, high-loss examples (e.g., cracks) are favored for the crack detection problem. A large body of work in computer vision-based methods for automated crack detection has been undertaken in recent years to address these challenges.

1.1. Related surveys on crack detection

To date, various techniques have been proposed and investigated for crack (defect, distress, damage, anomaly) detection and physical condition assessment of civil infrastructures. Several surveys have also been conducted to review the existing image-based automatic crack detection approaches from various perspectives. These surveys typically focus only on techniques for specific infrastructures or materials.

For example, some surveys focused on reviewing existing methods for specific infrastructures, e.g., pavements, tunnels, etc. The performance of six pavement distress segmentation methods was quantitatively evaluated in Tsai et al. (2009), including dynamic optimization, Canny edge, statistical thresholding, multi-scale wavelet, seed verification, and iterative clipping, by adopting a scoring measure. The work in Chambon and Moliard (2011) also focused on pavement assessment and compared the automatic detection methods that are based on morphological tools. These surveys only considered the traditional image processing methods. Wang (2011) discussed technological advances in automated detection of pavements using 3D laser imaging methodology. Instead of reviewing detection methods, they focused on the technical details and difficulties in capturing 1-millimeter surface images of pavements and introducing their 3D laser imaging system called PaveVision3D. Recently, the work in Hsieh and Tsai (2020)

reviewed techniques for processing 3D pavement images and evaluated the performance of shallow machine learning and deep learning algorithms. The authors also suggested that solving the false-positive problem is an important direction in further work.

On the other hand, some other surveys focus on methods that detect cracks on surfaces of a particular material. For example, Koch et al. (2014) reviewed earlier vision-based inspection methods for large concrete structures using Building Information Modeling (BIM) techniques. Payab et al. (2019) briefly conducted the development using image-based methods for detecting cracks in concrete surfaces. However, these works assumed that cracks propagate inherently with linear relations and only considered morphological-based methods. Jahanshahi et al. (2009) reviewed image-based approaches for automatic detection of cracks and corrosion in bridge structures of steel material. Neogi et al. (2014) reviewed defect detection and classification algorithms for steel surfaces. Zakeri et al. (2017) focused on asphalt pavement and reviewed image processing approaches for crack detection using various data acquisition platforms (e.g., Digital Highway Data Vehicle (DHDV), robotic system, and Quadcopter Unmanned Aerial Vehicle (QUAV/UAV) system). Koch et al. (2015) also reviewed vision-based defect detection and condition assessment methods for civil infrastructures of multiple materials. They considered materials of both concrete and asphalt, and discussed algorithms for several types of distresses. The survey paid less attention to crack detection and analysis. Recently, Czimmermann et al. (2020) reviewed defect detection for several materials, including metals, ceramics, and textiles, with 13 kinds of defects such as scratch, corrosion, galling, color error, etc.

In addition, the work in Mohan and Poobal (2018) reviewed crack detection techniques of 50 articles. The authors discussed various types of images that have been used for crack detection. They also presented taxonomies of existing methods based on various factors for analysis, such as detection objective, accuracy/error level, and image processing techniques. However, the authors only reviewed traditional methods based on low-level features of cracks. In addition to crack detection algorithms, it reports several state-of-the-art developments in the Pavement Asset Management System (PAMS) in the work of Peraka Naga Siva Pavani (2020). They mainly focus on pavement image acquisition and defect assessment techniques rather than crack detection techniques.

Next, we will describe the scope of this survey and contrast it with the related surveys discussed in this subsection.

1.2. Scope of the survey

Crack detection has broad applicability in assessing various civil infrastructures, such as tunnels, walls, buildings, dams, bridges, beams, pavement, etc. In addition, civil infrastructures are typically constructed using three types of materials, i.e., asphalt, concrete, and metal-like.

Due to different construction materials and usage environments, the cracks vary significantly in shapes/width, continuity, backgrounds, noises, lightning conditions, textures, etc. There are also many crack types: longitudinal, transverse, and complex (alligator, block, mesh, diagonal). In this survey, we do not limit our scope to any specific infrastructure, and we review crack detection of infrastructures that are built from the following materials.

Asphalt has been used in many infrastructures like buildings, dams, and especially in the construction of pavements. Asphalt pavements are often subjected to large crowds, frequently in poor condition, chronically underfunded, and are becoming increasingly hazardous. It is estimated that 1/5 of the highway pavements are in poor condition, and there is a significant and increasing backlog of rehabilitation needs. Cracks are the most common and critical defects of pavements which are the primary indicators to reflect the status of the deterioration pattern. Detecting pavement cracks is important to facilitate road maintenance, provide a better experience in automatic driving, and reduce the risk of accidents (e.g., collisions, falls).

Concrete is not as strong and tough as steel but has many excellent characteristics, e.g., resistance to fire and water, corrosion, cyclic loading, etc. Thus, concrete material has been widely used in many infrastructures including bridges, pavement, tunnels, dams, piles, foundations, footing, floors, beams, columns, roofs, exterior walls, pipes, sewage systems, etc. The cracking tendency is one of the most serious deterioration compared to other defects, which is a major cause of concrete failures due to their low tensile strength. Crack is one of the most important indicators to characterize the safety of concrete structures. For example, it is used as one of the few quantitative indicators specified in the bridge maintenance regulations. Monitoring, prevention, and control of cracks on pavement surfaces play a very important role in ensuring the safety and normal operation of public transportation.

Metal-like material, e.g., steel, is ubiquitous due to its resistance to natural and artificial wear. They are among the most important building materials for modern city construction. They have been widely used in constructing infrastructures like wind turbine blades, nuclear power plants, bridges, slabs, hot/cold strips, aircraft structures, etc. Typically, metal-like material could yield defects not only during the forming processes but also during the service period. The changes in the surface characteristics of steel over its service life promote the ingress of dissolved contaminants. Thus, it is vital to monitor those variables continually. The common defects of metal-like material are scratch, scarfing, and cracks. Cracks often appear at the critical structural joints of steel bridges that are vulnerable to fatigue, and grow in depth, length, as well as quantity. Detection of cracks is essential to (1) improve the yield of metal products during processing and production (surface-quality control); (2) monitor the safety status of metal-like material based infrastructures.

Images of infrastructures of different materials vary significantly in illumination conditions and surface textures (Chambon and Moliard, 2011; Koch et al., 2015). In contrast, concrete cracks usually have good continuity with relatively regular width, due to the unique characteristics of external forces (Li et al., 2019b). Also, concrete cracks show obvious texture differences from the backgrounds even when the background is smeared. On the other hand, cracks of metal-like material are typically thin and small (with a high length/width ratio). They usually appear similar to scratches, making them challenging to detect correctly (Jahanshahi et al., 2009; Tabernik et al., 2019). In many infrastructures, pavement cracks and concrete cracks could co-exist and affect each other.

This survey will cover computer vision-based crack detection on all infrastructures constructed using the three materials discussed above. Different from the existing surveys, we also review techniques that have been used for data acquisition and present publicly available datasets that are catered to different application scenarios. We will discuss the intuition and underlying assumptions of the existing methods, their key ideas and theoretical foundation, and the advantages and disadvantages. We do not compare the absolute performance of the reviewed methods as they used the different experimental setups (e.g., data sources, evaluation metrics, model configurations). However, we will identify and discuss the common techniques and challenges of the studies, and propose potential future directions to further advance the field.

1.3. Contributions and outline

Compared to existing surveys, we provide a more comprehensive review of vision-based crack detection, by covering multiple infrastructure materials (asphalt, concrete, metal-like), various components of a typical computer vision framework (including data acquisition, public datasets, algorithms, evaluations), and discuss unsolved issues and future directions. The contributions of this survey can be summarized as follow: (1) We discuss and compare widely-used data acquisition techniques for crack detection with regard to different application scenarios. We also summarize up-to-date popular datasets that are publicly

available. (2) We provide a comprehensive overview of the existing detection algorithms for crack image analysis, from traditional image processing algorithms and classical machine learning-based methods to state-of-the-art deep-learning methods. We show a Sankey diagram of the existing works addressing crack detection problems, which provide insights to efforts made regarding various infrastructure materials and detection methodologies. Then, we introduce a method-based taxonomy and review representative algorithms of each category in a systematic manner. (3) We introduce and discuss various standard evaluation metrics employed in the literature. (4) Finally, we discuss the unsolved issues in crack detection and provide several promising directions for future works to advance this research field.

The rest of this review is organized as follows: Section 2 describes the widely adopted image and video acquisition techniques for civil infrastructures and publicly available crack datasets. In Section 3, the widely used evaluation metrics are discussed. Section 4 first provides a taxonomy of crack detection methods proposed over the past two decades, including threshold-based methods, morphology methods, shallow machine learning methods, and deep learning methods. Then, each category is discussed in the subsequent subsections. Section 5 discusses the unsolved issue in this field and promising directions for future work. Section 6 concludes the paper.

2. Acquisition techniques and public datasets

2.1. Acquisition techniques

The crack detection problem is a challenging class-imbalance problem due to a small proportion of cracks. Thus, it is critical to capture high-quality images of the target structures, which helps to produce accurate ground truths to facilitate crack detection model training. During the past few decades, many non-destructive vision detection techniques have been proposed and developed.

As a contactless and cost-effective acquisition method, industrial/commercial cameras are easy to deploy and capable of providing different resolutions of output images/videos. Thus, they are widely used for monitoring various target infrastructure surfaces. The 2D cameras have the drawbacks of being sensitive to varying lighting conditions, leading to low-intensity contrast between cracks and surrounding textures, which negatively impacts the subsequent detection performance. To mitigate such impacts, pre-processing techniques are required, which increases the computation overheads. With the progress of optical technique and memory chip, the 3D laser imaging technique has become a promising and popular approach for automatic acquiring surface images of pavements, tunnels, etc. Compared to the traditional 2D crack image method, a 3D laser scanning system can better characterize cracks in case of shadows and tire marks. 3D laser scanning systems can capture more useful information even with poor lighting conditions. In addition, compared with a 2D camera, the 3D laser can capture depth information of cracks which provides extra indicators to evaluate the structure status. However, imaging using a 3D laser produces a significant amount of scanning noise in addition to the high costs. Also, 3D laser imaging methods produce a large amount of point cloud data with noise and format issues, which require a lot of effort to process the raw data.

In addition to the 3D laser scanning system, the depth perception of an RGB-D camera can also provide extra information (i.e., distance to the objects) for estimating the size of an object. To improve crack detection performance relying on depth formation, Jahanshahi et al. proposed 3D scene reconstruction techniques using a traditional RBG camera (Jahanshahi and Masri, 2012; Jahanshahi et al., 2013) and an inexpensive RGB-D camera (Jahanshahi et al., 2012), respectively. They captured images from different angles for each scene and obtained the depth information by scaling the reconstructed 3D model of a scene. Then, feature extractors and classifiers are employed to extract cracks on civil infrastructure surfaces. They also used an inexpensive RGB-D sensor to capture the pavement images, and an unsupervised crack detection algorithm based on Random Sample Consensus (RANSAC) was proposed. Yang et al. (2015) captured 3D images of a reinforced concrete (RC) structural by employing stereo triangulation with two high-resolution cameras. They proposed a full procedure for unfolding the two regions using image rectification techniques. They also developed a program named ImPro Stereo based on MATLAB GUI and OpenCV library to visualize the final results. The in-situ bridge cyclic test results showed that the system could detect cracks with 0.2 pixels in photos.

In addition, there are some intelligent platforms that work autonomously, have low cost, strong reliability and high performance. Instead of fixing the position of cameras, some intelligent platforms have been developed to enable crack imaging using moving cameras that operate with low cost, strong reliability, and high performance. The platforms in existing works for capturing crack images can be generally grouped into three categories: handheld, vehicle-mounted, and UAV-mounted.

In the first category, smartphones are the most common handheld devices. For instance, Shi et al. (2016) built an open pavement crack dataset using iPhone5. Kalfarisi et al. (2020) proposed a dataset that was built during the field inspection process, where the images are captured using handheld cameras by professional inspectors. Unfortunately, the dataset is not made publicly available. The handheld method is convenient for data collection in some scenarios, but the scale of the resulting dataset is limited due to low efficiency. Moreover, the quality of the collected data is not guaranteed due to factors such as uneven lighting conditions.

Compared to handheld methods, vehicle-mounted crack imaging systems can provide uniform lighting conditions for capturing crack images/videos. An inspection vehicle can combine various equipment to detect distress and collect multiple interesting results in time. To date, there are several commercial all-in-one solutions like Dyntest, Pathway, and the Applus System (Laurent et al., 2012). The work in Velinsky et al. (2003) developed a vehicle-based highway crack-sealing machine (using a CCD camera with area scan), which captures 2D data/images to evaluate the pavement conditions. The Road Monitoring and Assessment (RMA) system (Zakeri et al., 2013), designed by the German Road and Transportation Research Association (FGSV), can perform multiple tasks, including pavement image acquisition, condition analysis, and road quality assessment. Zhang et al. (2017b) employed a digital highway data vehicle fabricated by Pathway Corporation to capture crack images and proposed several crack detection neural networks, e.g., crackNet (Zhang et al., 2017c), CrackNet-V (Fei et al., 2019), and CrackNet-R (Zhang et al., 2019c). The vehicle can capture 1 mm 3D pavement image data at the sample rate of 100 KPH, with full coverage for a 4 m wide lane at a 30 kHz scanning rate. Vehicle-mounted imaging systems are also widely used for the inspection of tunnel walls. For example, the Laser Tunnel Scanning System (LTSS), designed by Pavemetrics,[1] can scan a full tunnel vault (with height up to 24 m) at a 1 mm resolution image and acquire 3D data at speeds up to 20 km/h. In addition to human-driven vehicles for surface inspection, as mentioned above, many researchers and engineers have developed robots to achieve automatic assessment process (Chotiprayanakul et al., 2012; Guo et al., 2009; Tseng et al., 2011; Chu et al., 2013; Prasanna et al., 2014). The robots can increase safety and convenience during the data acquisition process but lead to extra challenges of path planning and automatic control of the robots. In general, vehicle-mounted systems are expensive and could cause significant redundancy in the obtained dataset. Also, inspection vehicles typically are of large size, as shown in Fig. 2(b)–(d), which limits their application scenarios.

[1] http://www.pavemetrics.com.

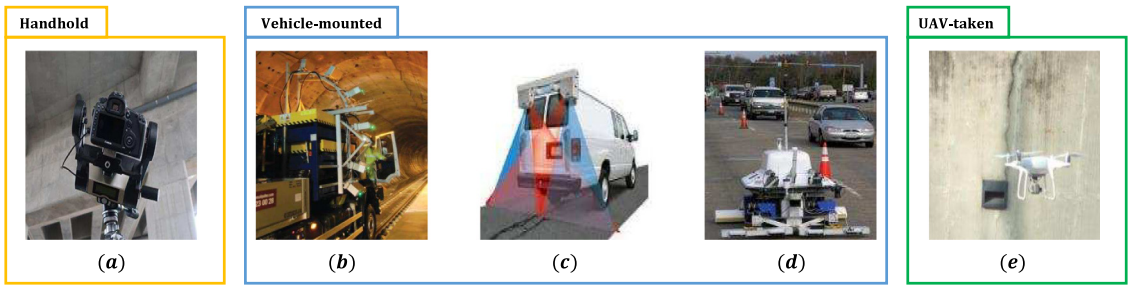

Fig. 2. An illustration of acquisition device for civil infrastructure surface images. (a) Canon DSLR full-frame camera to capture a bridge image; (b) Laser Tunnel Scanning System (LTSS) developed by pavemetrics. Co.Ltd for automated inspection of transportation infrastructures. It can achieve 1 mm resolution 3D image data with 20 km/h; (c) LRIS system developed by INO.Co.Ltd can capture pavement image data with two line scan cameras at 100 km/h; (d) Robotic Assessment Bridge Inspection Tool for imaging bridge decks developed by Federal Highway Administration; (e) A Phantom 4 advanced UAV to detect cracks on concrete building.

Table 1
Popular public crack detection datasets[a].

Name	Material	No. samples	Resolution	Device	Pixel-labeled	Metrics
KolektorSDD[2]	Metal-like	400	1408 × 512	Industrial camera	√	$AP\&(FP+FN)$
DeepCrack[3]	Hybrid	537	544 × 384	–	√	$Acc\&mIoU\&Pr\&Re\&F1\&AUC$
Concrete-Crack-Ozgene[4]	Concrete	40 000	227 × 227	–	×	$AP\&F1$
CFD[5]	Pavement	320	480 × 320	iPhone5	√	$Pr\&Re\&F1\&CI$
CrackTree[6]	Pavement	206	600 × 800	Area-array camera	×	$Pr\&Re\&F1$
Bridge-Crack-Xu[7]	Concrete	2068	1024 × 1024	Phantom 4 camera	√	$Acc\&Pr\&Re\&F1\&TNR$
Pavement-Crack-Chambon[8]	Pavement	38	Multi	–	√	$SEN\&SPC\&DSC$
TITS[9]	Pavement	269	Multi	2D laser	√	$Pr\&SEN\&DSC$
Crack500[10]	Pavement	500	360 × 640	Mobile phone	√	$AIU\&ODS\&OIS$
GAPS384[11]	Pavement	1969	1920 × 1080	2D laser	√	$F1\&BER$
SDNET2018[12]	Hybrid	230	256 × 256	Nikon camera	×	–
Pavement-Crack-Maeda[13]	Pavement	9053	600 × 600	LG-5X	×	$Pr\&Re\&IoU$
EdmCrack600[14]	Pavement	600	1920 × 1080	GoPro 7	√	$Pr\&Re\&F1$
Severstal[15]	Metal-like	14 396	256 × 1600	–	√	DSC
LRRD[16]	Hybrid	52 408	256 × 256	3D laser	×	$Pr\&Re\&F1$

[a] AP (Average Precision), FP (False Positives), FN (False Negatives), Acc (Accuracy), $mIoU$ (mean Intersection over Union), Pr (Precision), Re (Recall), $F1$ (F_1 $score$), AUC (Area Under the Curve), CI (Continuity Index), TNR (True Negative Rate), SEN (Sensitivity), SPC (Specificity), DSC (Sørensen–Dice Similarity Coefficient), AIU (Average Intersection over Union), ODS (Optimal Dataset Scale), OIS (Optimal Image Scale), BER (Balanced Error Rate), IoU (Intersection over Union). Please refer to Section 3 for the specific definitions of the above metrics.

In some scenarios, the vehicle-mounted inspection systems are still not efficient enough, especially when a variety of inspection equipment needs to be installed. This motivates the usage of QUAV/UAV imaging technology that can detect bridge/tunnel cracks more conveniently without affecting the normal operation of the bridge/tunnel. For example, a UAV equipped with camera equipment as well as a variety of sensors can easily capture images of the crack/distress on the surface of the structure without affecting its functions. In addition, the use of QUAV/UAV is also driven by the requirement of collecting high-quality images for special application environments, e.g., bridge (Yan et al., 2021), high wall (Zhu et al., 2018), large-scale wind turbine (Peng and Liu, 2018; Wang and Zhang, 2017), concrete wall (Kim et al., 2017b), and building facades (Pereira and Pereira, 2015), etc. However, the imaging quality of UAVs may not meet the detection requirements for certain applications due to factors such as airflow fluctuations during the flight, sunshine direction, wind speed, weather conditions, etc. In addition, the deployment feasibility and efficiency of UAV-mounted imaging systems are also affected by many other factors, including (1) the continuous operation time relying on lithium batteries; (2) the mode accuracy of UAV positioning and navigation; (3) obstacle avoidance robustness in UAV operation, etc.

2.2. Public datasets

A major problem in crack detection studies is the lack of standard benchmark datasets. Benchmark datasets are critical for research as findings can be easily compared and integrated, whereas the lack of benchmark datasets could result in data-specific rather than generalizable methods. As such, we summarize popular and public datasets which are often used for crack detection tasks. Dataset availability was determined at the time of writing. Most of these datasets have provided appropriate pixel-wise labeling. Table 1 illustrates the popular public crack detection datasets, where the download link of each dataset has been provided. Fig. 3 shows the example image data for the public datasets listed in the above table. The following section describes the details of each dataset.

KolektorSDD,[2] which is managed by Tabernik et al. (2019), is a large-scale public dataset containing 400 images of defected electrical commutators. Each image was captured in a controlled environment with a resolution of 1408 × 512 pixels. The dataset was split into three groups to perform a 3-fold cross-validation. It is worth noting that crack is a primary kind of defect included in it. The dataset is used to train and test the algorithm to identify complex surface defects on metal-like surfaces for surface-quality control. The dataset is publicly available to boost the development and evaluation of new methods for surface-defect detection. Example images are shown in Fig. 3(a).

DeepCrack.[3] Liu et al. (2019c) created a crack dataset of concrete/pavement images. The dataset contains 537 RGB color images with a resolution of 544 × 384 pixels. Each image contains a single or multiple cracks, which have been labeled at the pixel-wise level. The dataset contains crack images in multi-scale and multi-scene to represent 261

[2] http://www.vicos.si/Downloads/KolektorSDD.
[3] https://github.com/yhlleo/DeepCrack.

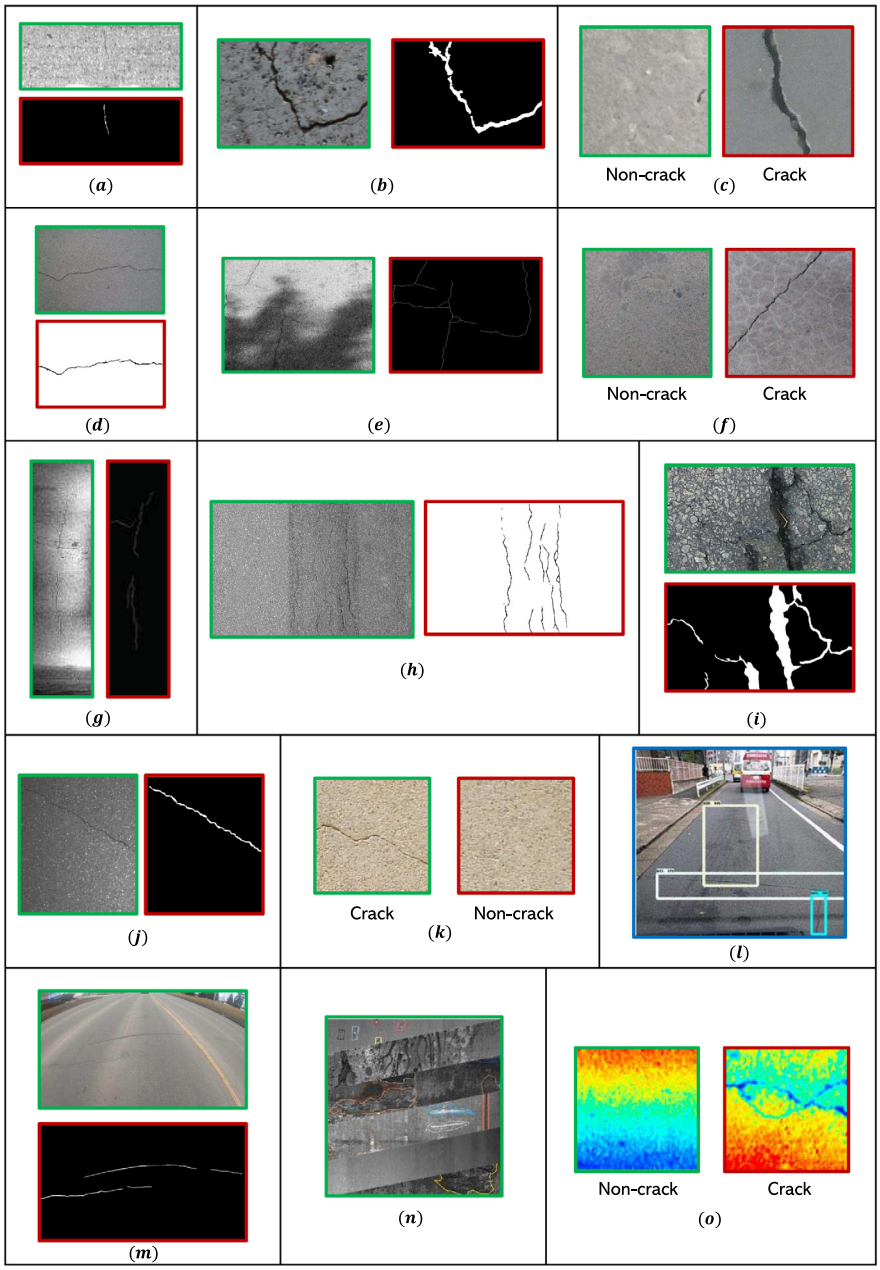

Fig. 3. Examples of crack images in different public datasets.

characteristics of cracks universally. Example images are shown in Fig. 3(b).

Concrete-Crack-Ozgene.[4] Özgenel and Sorguç (2018) published a large-scale concrete dataset that contains 40K image patches with a resolution of 227 × 227 pixels. It includes both positive and negative samples, where a positive sample contains cracks while a negative sample contains no cracks. Specifically, these image patches are extracted from large-resolution concrete building images that are captured by fixed-distance (>1 m) cameras under similar illumination conditions, as shown in Fig. 3(c).

CFD.[5] Shi et al. (2016) built a CrackForest-Dataset (CFD), which has been one of the most popular pavement crack datasets in relative studies. The dataset comprises 118 pavement images that are captured by a mobile phone (with a resolution of 480 × 320 pixels), as shown in Fig. 3(d). The images of this dataset contain a significant amount of noisy pixels, such as oil spots and water stains. Many recent crack detection algorithms have evaluated their methods based on this dataset.

CrackTree.[6] Zou et al. (2012) collected another popular pavement crack dataset that consists of 260 pavement images that are captured by an area-array camera, as shown in Fig. 3(e). Compared with dataset CFD[5], its images have a high resolution of 800 × 600 pixels but suffer from quality problems. For example, a large portion of images contain shadows that cause difficulties for crack detection. The authors

[4] https://data.mendeley.com/datasets/5y9wdsg2zt/2.
[5] https://github.com/cuilimeng/CrackForest-dataset.
[6] https://sites.google.com/site/qinzoucn.

proposed a shadow-removal algorithm as a pre-processing stage to improve the data quality.

Bridge-Crack-Xu.[7] Xu et al. (2019) created a patch-based crack dataset of bridge images using a Phantom 4 camera. The original dataset consists of 2068 bridge images with a resolution of 1024 × 1024 pixels, as shown in Fig. 3(f). In order to generate more samples for training crack detection models, data augmentation techniques such as random cropping and random flipping are employed in the original dataset. The resultant dataset contains 6069 samples (4856 training samples and 1213 testing samples in their experiments).

Pavement-Crack-Chambon.[8] Chambon and Moliard (2011) published a dataset of pavement cracks in 2011, which are captured by an acquisition system consisting of 4 video cameras, as shown in Fig. 3(g). The dataset contains image data of three categories: difficile, dynamique, and statique. The dataset has been used to test algorithms for extracting cracks with different qualities of the road surface or road texture. However, this dataset is small scale consisting of only 38 images.

TITS.[9] Amhaz et al. (2016) published a crack dataset that is composed of 269 real images and one synthetic image. The images of the dataset were captured by different kinds of optical sensors, including Aigle-RN(62), ESAR(30), LCMS(65), LRIS(89), and TMEPEST2(23), as shown in Fig. 3(h). Among those sensing systems, only ESAR corresponds to a static acquisition (not mounted on a vehicle) without controlled lighting conditions. In contrast, the other four are dynamic ones (mounted on a vehicle) with controlled lighting conditions. The dataset has been used to test the behavior of the pavement crack detection algorithms in a realistic context.

Crack500.[10] Yang et al. (2019) created a public pavement crack dataset that contains 500 images with a resolution of 2000 × 1500 pixels, as shown in Fig. 3(i). The images were captured by mobile phones, and each image was manually labeled in pixel-wise granularity.

GAPS384.[11] Eisenbach et al. (2017) proposed a public dataset consisting of images of German Asphalt Pavement Distress (GAPs). It has a total of 1969 gray images with a resolution of 1920 × 1080 pixels, as shown in Fig. 3(j). In addition to images of cracks, the dataset also contains many pavement distress images, e.g., potholes, patches, and joints. Each image in the dataset is labeled in pixel-wise granularity.

SDNET2018[12] is a dataset of concrete crack images that are captured from 54 bridge decks, 72 walls, and 104 pavements using a COOLPIX L830 Nikon digital camera (Dorafshan and Maguire, 2017). The original images were segmented into over 56 000 image patches of crack and non-crack concrete sub-images, where each sub-image has a resolution of 256 × 256 pixels, as shown in Fig. 3(k). In general, the width of cracks in this dataset ranges from 0.06 mm to 25 mm.

Pavement-Crack-Maeda[13] is a large-scale pavement condition dataset consisting of images of road damages, as shown in Fig. 3(l). The images were captured by a car-mounted smartphone moving at a speed of about 40 km/h. The dataset is annotated manually in the manner of PASCAL VOC (Everingham et al., 2015). In summary, the dataset contains 9053 damage images with a resolution of 600 × 600 pixels (Maeda et al., 2018), and there are 15 435 damage bounding boxes in total.

EdmCrack600[14] is a dataset consisting of 600 images captured by more than 20 videos GoPro Hero 7 device (Mei and Gül, 2019), as shown in Fig. 3(m). The images have a resolution of 1920 × 1080 pixels and are captured under various parameter factors like illumination variation, noise, and interruption. This dataset is presently the largest crack dataset of pavement images annotated at pixel level among all the public datasets. In addition to common environmental noises like shadows and occlusions, the dataset contains many non-pavement region noises like cars, lane makers, houses, and the sky.

Severstal[15] is a steel defect dataset published by Kaggle (a website for data science competitions). In this competition, researchers need to predict the location and type of defects found in steel manufacturing. Similar to KolektorSDD[2], crack is a primary kind of defect included in the dataset. It contains 14 369 images (with a resolution of 256 × 1600 pixels), where 12 568 images are used for training and 1801 images are used for testing in the defect detection competition. An image could contain no defect, defects of a single class, or defects of multiple classes, as shown in Fig. 3(n).

LRRD[16] is a dataset of laser-scanned range images captured from asphalt and concrete roadways. The images are captured by a laser image and measurement system, which is manufactured by AMES Engineering (Zhou and Song, 2020). Specifically, there are 25 785 crack samples and 26 623 non-crack samples with a resolution of 256 × 256 pixels, as shown in Fig. 3(o). Different from crack samples, the non-crack samples possess different patterns, e.g., edges, grooves, clean surfaces, shoulder drop-offs, etc.

Table 2
Confusion matrix of a crack detection task.

Total = P + N	Predicted crack (PP)	Predicted non-crack (PN)
Actual crack (P)	TP	FN
Actual non-crack (N)	FP	TN

3. Evaluation metrics

To assess the performance of crack detection algorithms, many evaluation metrics have been proposed based on the confusion matrix as shown in Table 2. The matrix contains two rows and two columns and reports the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), characterizing the results of a crack detection algorithm.

Correctly labeling the crack boundary pixels is challenging, and the unreliable labels of boundary pixels have a negative impact on the measurement of accuracy. To reduce this impact, many works adopt a technique called tolerance margin to calculate detection accuracy. Specifically, a pixel that is detected as a crack and located at most β-pixels ($\beta = 1$ to 5) away from a real crack pixel will be considered a TP case. This technique pays slighter attention to boundary pixels, which enables mitigating the influence of some unreliable labels.

Many evaluation metrics have been proposed to reflect the performance of crack detection algorithms in terms of accuracy. Among them, the Accuracy (Acc), Precision (Pr), Recall (Re), and $F_1 - score$ are most popular for evaluating the performance of crack detection algorithms. It should be noted that Recall is also denoted as True Positive Rate (TPR) or Sensitivity (SEN).

$$Accuracy(Acc) = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision(Pr) = \frac{TP}{TP + FP} \tag{2}$$

$$Recall(Re) = \frac{TP}{TP + FN} \tag{3}$$

$$F1 - score(F1) = \frac{2 \times Pr \times Re}{Pr + Re} \tag{4}$$

[7] https://github.com/tjdxxhy/crack-detection.
[8] http://perso.lcpc.fr/sylvie.chambon/FISSURES/Datasets.html.
[9] https://www.irit.fr/~Sylvie.Chambon.
[10] https://github.com/fyangneil/pavement-crack-detection.
[11] https://www.tu-ilmenau.de/neurob/data-sets-code/gaps.
[12] https://digitalcommons.usu.edu/all_datasets/48/.
[13] https://github.com/sekilab/RoadDamageDetector.
[14] https://github.com/mqp2259/EdmCrack600.
[15] https://www.kaggle.com/c/severstal-steel-defect-detection/overview.
[16] https://www.designsafe-ci.org/data.

In contrast, some metrics are proposed to characterize detection errors that indirectly reflect the accuracy, including True Negative Rate (TNR), Specificity (SPC), False Negative Rate (FNR), Balanced Accuracy (BA), Balanced Error Rate (BER), Positive Predictive Value (PPV), Negative Predictive Value (NPV), False Omission Rate (FOR), False Discovery Rate (FDR).

$$\begin{aligned} TNR &= \frac{TN}{TN+FP}, \quad FNR = \frac{FN}{TP+FN} \\ PPV &= \frac{TP}{TP+FP}, \quad NPV = \frac{TN}{TN+FN} \\ FOR &= \frac{FN}{TN+FN}, \quad FDR = \frac{FP}{TP+FP} \\ BA &= \frac{TPR+TNR}{2}, \quad BER = 1 - \frac{TPR+TNR}{2} \end{aligned} \tag{5}$$

In addition to the above-mentioned metrics that characterize the detection accuracy, many metrics have been developed to reflect various properties of the detection performance. For example, the Receiver Operating Characteristic (ROC) curve and the Precision–Recall (PR) curve are widely adopted to show a comprehensive comparison of the crack detection results and provide more intuitive comparisons.

The confusion matrix (Table 2) can be used to construct a point in either ROC-curve or PR-curve. In the ROC-curve, one plots the FPR on the x-axis and the TPR on the y-axis, where the FPR measures the fraction of negative examples that are misclassified as positive while the TPR measures the fraction of positive examples that are correctly labeled. In PR-curve, Recall is on the x-axis and Precision on the y-axis, whereas Precision measures the fraction of examples classified as positive that are truly positive. For the ROC-curve, the Area Under the Curve (AUC) is usually computed using the trapezoidal rule in a ROC-curve and provides an aggregate measure of performance across all possible classification thresholds. The Average Precision (AP) is employed to summarize a PR-curve as the weighted mean of Precision(s) achieved at each threshold. In general, computing the average AP of all classes of cracks leads to the mean Average Precision (mAP)

Compared to metrics such as the subset accuracy, the F1 score, ROC-curve, and PR-curve do not require optimizing a threshold for each label. The values of AUC and AP are between 0 and 1, where higher is better. Both the ROC-curve and PR-curve are commonly used tools that illustrate the performance of a binary classifier concerning all possible thresholds. It has been reported that when dealing with highly skewed datasets, the PR-curve exhibits a more informative picture of an algorithm's performance (Davis and Goadrich, 2006).

Moreover, the Intersection over Union (IoU) is one of the most widely used metrics in crack location and segmentation applications. IoU, also known as the Jaccard index, is widely used for common semantic segmentation and scene parsing evaluations (Long et al., 2015). Similar to mAP, the mean IoU ($mIoU$) is computed as the average of the IoU of each class. Frequency Weighted IoU ($FWIoU$) improves on $mIoU$ by assigning weights to each class based on its frequency of occurrence.

$$mIoU = \frac{1}{N}\sum_{i=1}^{N}\frac{n_{ii}}{t_i + \sum_{j=1}^{N} n_{ji} - n_{ii}} \tag{6}$$

$$FWIoU = \left(\sum_{k=1}^{N} t_k\right)^{-1}\sum_{i=1}^{N}\frac{t_i n_{ii}}{t_i + \sum_{j=1}^{N} n_{ji} - n_{ii}} \tag{7}$$

In some works of crack detection (Kalfarisi et al., 2020; Chambon and Moliard, 2011; Amhaz et al., 2016), IoU is also denoted by DSC (Sørensen–Dice Similarity Coefficient) and used to compare the pixel-wise agreement between a predicted segmentation, and its corresponding ground truth based on the following formula:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|} \tag{8}$$

where $|X|$ is the predicted set of pixels and $|Y|$ is the ground truth. IoU computes the area of overlap between the predicted bounding box and the ground-truth bounding box, and is used as a metric to evaluate how close the prediction bounding box is to the ground truth. The IoU has the appealing property of scale invariance. An IoU score > 0.5 is normally considered a "good" prediction. A larger IoU means that the predicted result is closer to the true value.

Recently, Average Intersection over Union (AIU) was proposed by Yang et al. (2019) to evaluate the performance of crack detection algorithms without non-max suppression (NMS) and thinning operation. Compared with IoU, AIU takes the object width information into consideration; therefore, AIU can be used to determine if the width is precisely estimated, which is useful to assess the crack (damage) degree of pavement condition.

$$AIU = \frac{1}{N_t}\sum_{t}\frac{N_{pg}^t}{N_p^t + N_g^t - N_{pg}^t} \tag{9}$$

where t is a threshold, N_t denotes the total number of thresholds, N_p^t, N_g^t, and N_{pg}^t denote the number of pixels of predicted, ground truth, and their intersected crack region, respectively. Moreover, since the crack is similar to the edge/contours, they also adopted the Optimal Dataset Scale (ODS) and Optimal Image Scale (OIS) to further evaluate the effectiveness of crack detection algorithms.

$$\begin{aligned} ODS &= \max\left(2P_t \times R_t / \left(P_t + R_t\right)\right) \\ OIS &= 1/N_{img} \times \sum_{i}^{Nimg} \max\left(2P_t^i \times R_t^i / \left(P_t^i + R_t^i\right)\right) \end{aligned} \tag{10}$$

where N_{img} is the total number of images and t is a threshold ($N_{img} = 0.01, 0.02, \ldots, 0.99$).

A Scoring Measure (SM) based on the buffered Hausdorff distance (HD-score) between two images was proposed to demonstrate the performance of crack detection algorithms (Tsai et al., 2009; Jiang and Tsai, 2015; Zhang et al., 2020). The metric is insensitive to the detection task with highly imbalanced foreground-background. Recently, the work in Zhang et al. (2020) demonstrated that IoU is not suitable for evaluating crack detection algorithms since cracks usually exhibit a long-narrow target and occupies a tiny proportion of an image. The score has a value ranging from 0 to 100, where a score of 100 indicates the most accurate results. The score can reflect the overall crack localization accuracy.

$$\mathrm{SM} = 100 - \frac{\mathrm{BH}(A,B)}{L} \times 100 \tag{11}$$

$$\mathrm{BH}(A,B) = \max[h(A,B), h(B,A)] \tag{12}$$

$$h(A,B) = \frac{1}{m}\sum_{a \in A} \mathrm{sat}_{\mathrm{L}} \min_{\mathrm{b} \in \mathrm{B}} \|a - b\| \tag{13}$$

where A and B are the matrices of crack locations in the ground truth and predicted image, and $BH(A,B)$ measures the distance between image A and B. Let L be the saturation constant, the operation sat_L is defined as: if the distance between a crack pixel and its closest crack pixel in the other image exceeds L, then, sat_L is used to indicate the distance; otherwise, sat_L is used the difference of these two pixels.

Moreover, the work in Sun et al. (2012) employed the waveform similarity to evaluate the consistency between actual and predicted crack shapes using:

$$R = \frac{\sum[x(t) \times y(t)]}{\sqrt{\sum x(t)^2 \times \sum y(t)^2}} \tag{14}$$

where $x(t)$ and $y(t)$ are the actual and predicted crack waveform. R ranges in [0,1].

4. Algorithms

Many existing surveys have proposed taxonomy methods for crack detection based on 'pre-processing → segmentation → feature extraction → detection' (Zakeri et al., 2017; Koch et al., 2015). In this

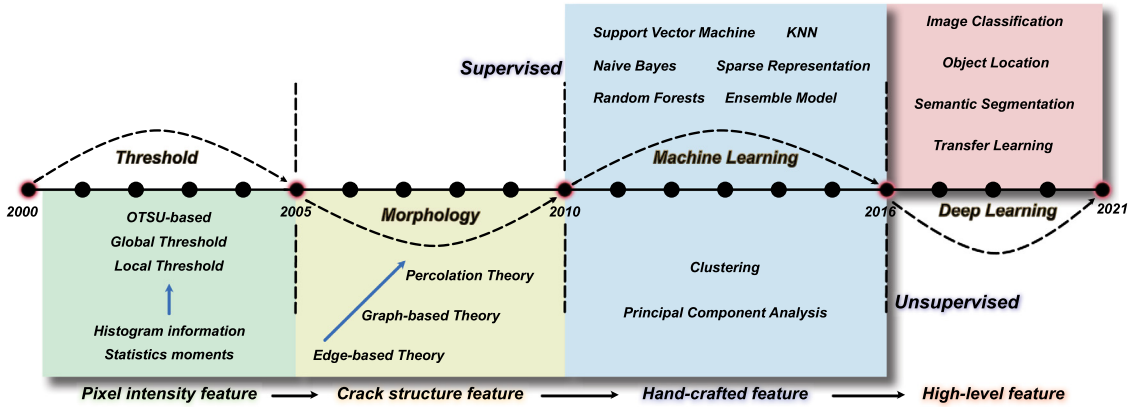

Fig. 4. Categorizing general computer vision based crack detection methods.

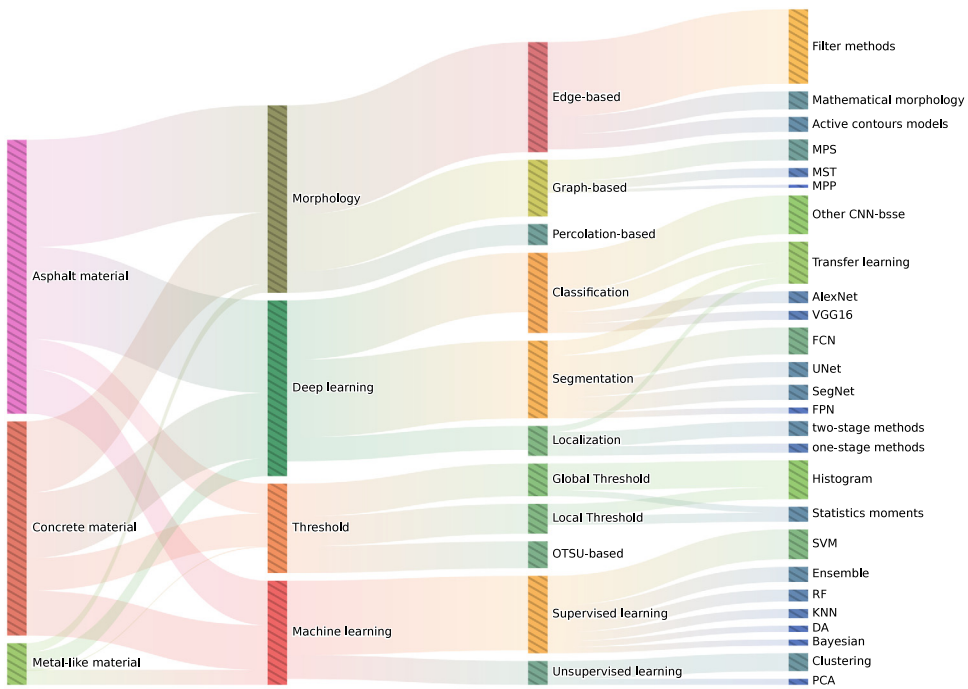

Fig. 5. The Sankey diagram of computer vision based manual/automatic inspection methods for civil infrastructures.

survey, we adopted a new taxonomy based on the development of automatic crack detection algorithms, as shown in Fig. 4. Generally, the threshold-based methods are widely used in the early stage (i.e., between 2000 and 2005), which classify crack and non-crack pixels solely based on pixel intensity information. Then, many works considered the morphology characteristics of cracks (e.g., topology, continuity, contour length, etc.) to develop more robust crack detection algorithms between 2005 and 2010. Subsequently, significant research efforts have been devoted to the development of machine learning-based detection algorithms to incorporate multiple high-level features for achieving better detection accuracy, including both supervised and unsupervised machine learning methods. Major efforts have been made on traditional shallow machine learning methods between 2010 and 2016, such as Support Vector Machine (SVM), Random Forest (RF). Due to the strong capabilities of capturing spatial–temporal dependencies, deep learning models have been widely employed in crack detection problems since 2016 and demonstrated promising performance of real-time, automatic, and robust crack detection. In general, the above methods used different feature information to characterize and detect cracks. Specifically, threshold-based methods are based solely on pixel intensity features, morphology methods utilize crack structure features, and shallow machine learning methods rely on hand-crafted features. Instead of using hand-crafted features or hard-coded knowledge that needs high domain expertise, deep learning methods learn invariant high-level features in an end-to-end manner.

Fig. 5 illustrates a Sankey diagram which shows a quantitative comparison of the existing efforts on crack detection with regards to different materials. The figure shows that, compared to metal-like material, more efforts have been made to crack detection on asphalt material and concrete material. Among the four categories of methods discussed earlier, morphology methods and deep learning methods attracted more attention. The former is widely used in years between 2005 and 2010, while the latter has gained popularity in recent years. It is expected that more deep learning-based efforts will be undertaken. For morphology methods, more efforts have been made to edge-based methods, followed by graph-based methods and percolation-based methods. The deep learning methods adopted different strategies for crack detection, including classification strategy, segmentation strategy, and localization strategy, where the first two strategies are more popular than the localization strategy. The threshold methods contain three subcategories: global threshold methods, local threshold methods, and OTSU-based methods. The shallow machine method general can be

divided into supervised learning and unsupervised learning, where the supervised learning methods are used more often. Fig. 5 shows a more detailed taxonomy of crack detection algorithms for each sub-category.

4.1. Threshold

The threshold-based methods assumed that the crack pixels are generally darker than the background or surroundings and thus have a lower gray intensity than other pixels. Thus, a threshold can be set up such that crack pixels have higher values than the threshold while non-crack pixels have lower values. In the past two decades, a variety of crack detection algorithms have been proposed, with novel techniques for selecting optimal thresholds, ranging from global to locally adaptive.

4.1.1. Global threshold

Various algorithms have been proposed to automatically find global and optimal threshold values over the past two decades. For example, Cheng et al. (2003) proposed a real-time image thresholding method for pavement crack detection. They calculated the mean and standard deviation of the pixel intensities to determine the threshold value for a gray-level pavement image. Tang and Gu (2013) computed the median and variance values of the histogram of the image to select the optimal threshold values. In addition, a Gaussian filter was adopted to smooth the histogram. Yu et al. (2019) proposed a low-high threshold method to detect cracks for concrete slope protection. The threshold is calculated based on the adjacency effect, which combines the dark target's location and intensity. They compared their method with the Canny algorithm and Stroke Width Transform (SWT) method and achieved better detection results. We refer the readers to works of Zhu et al. (2018), Lee et al. (2013), Kamaliardakani et al. (2014), Shen et al. (2015), Hoang (2018) and Liang et al. (2018) for other techniques that have been reported for setting optimal thresholds.

While calculating the thresholds from histograms works well for simple scenarios, it fails whenever the assumption that there are two peaks in the histogram is violated. This is because, the histogram of image data only reveals quantitative information of pixel intensity, neglecting their structural information. To overcome such limitations, many works utilized local information to set the thresholds, which exploits the crack information more effectively.

4.1.2. Local threshold

Cracks typically often appear in a small portion of a crack image, and there also could be various noisy pixels (e.g., due to background textures and lighting conditions during the data acquisition process), making it difficult to obtain a suitable threshold to distinguish crack from non-crack pixels. As such, many efforts have been made to determine the optimal threshold values based on local information accordingly.

Kirschke and Velinsky (1992) proposed a pavement crack detection approach by diving an image into a grid of tiles. Then, the characteristics of each tile are represented as histogram-based properties and statistical moments to determine a threshold value for making predictions. Li and Liu (2008) proposed a pavement image-thresholding algorithm based on the neighboring difference histogram method, which assumed that the thresholding value is strongly related to the standard deviation of pixels. Oliveira and Correia (2009a) proposed a dynamic thresholding method for pavement crack detection. They first used morphological filters in a pre-processing stage. Then, they divided a whole image into blocks of 75×75 pixels and utilized an entropy value as the indicator to determine if the block contains a crack. The work of Fujita and Hamamoto (2011) used a dynamic thresholding approach in a post-processing stage to the detected cracks at a fine-granularity level. A probabilistic relaxation technique was also utilized to consider the pixel correlation information, leading to accurate edges with good connectivity and less distortion. Kim et al. (2017b,a) compared five common segmentation methods for concrete crack detection, proposed a strategy of optimal parameter selection, and presented a crack detection strategy method based on statistical moments of a selected window for UAV-taken images. In summary, most of the above threshold crack detection algorithms use histograms and statistics moments of image blocks to calculate the optimal threshold values. However, the fixed and static method of determining threshold values still cannot produce satisfactory results for complex scenarios.

4.1.3. OTSU-based

The OTSU method was first proposed by Otsu (1979). It automatically determines a threshold value T by counting the histogram characteristics of the whole image or local block in the OTSU segmentation process. Since then, the OTSU-based methods have become a widely adopted thresholding method for crack detection algorithms.

The works of Lee et al. (2013) and Talab et al. (2016) have employed OTSU-based methods to determine suitable threshold values and detect cracks on concrete surfaces to address the challenges caused by noisy backgrounds. Shen et al. (2015) presented an OTSU-based crack detection method to detect cracks on tunnel video data collected from several wireless multimedia video sensors and relied on the OTSU-based thresholding strategy to adapt to the varying backgrounds of video frames.

Also, Zhu et al. (2018) presented a recursive OTSU algorithm of histogram thresholding for infrastructure crack segmentation. Although the OTSU threshold method depends on the variance between classes, some cracks cannot be correctly identified from the foreground class in some cases in their work. As such, they identify regions of interest (ROIs) from the original image and run the OTSU method only on the ROI that encloses a range of intensity containing crack features in every iteration and excludes the background region for thresholding.

Zhang et al. (2017a) developed an efficient pavement crack detection method based on a thresholding and region growing algorithm. In their work, an improved OTSU method is used as a pre-processing method to remove the sign markings such as oil stains, skid marks, shadows, and lanes based on the consideration that the intensities of sign markings are higher than that of road texture and the sign markings are smoother than road texture. Hoang (2018) developed an improved OTSU method to identify cracks on building surfaces by combining a Min–Max Gray Level Discrimination (M2GLD) method. In other words, the method consists of two parts: the M2GLD and the conventional OTSU algorithm. M2GLD is an image enhancement processing step that is applied before the OTSU method is applied for image thresholding.

Nisanth and Mathew (2014) developed a pavement crack detection method based on the OTSU method. Firstly, they calculated the dynamic threshold value using the OTSU method; then, they used the conventional OTSU value minus the standard deviation of image pixel intensities in their work to reduce the noise on pavement surface images. Kamaliardakani et al. (2014) develop a method to extract sealed cracks automatically on images of pavement surfaces. Their work uses three segmentation methods, i.e., the OTSU method, Kittler and Illingworth, and maximum triangle. However, it is reported that they are computationally expensive and need significant efforts of optimization to obtain the proper threshold value. Therefore, they developed a local minimum algorithm to determine the threshold heuristically. It is reported that the method can detect sealed cracks in different environments. Liang et al. (2018) employed a conventional OTSU method to segment concrete cracks from the background. However, in this work, the segmentation results always contain considerable noise. Thus, a denoising method based on the crack geometric characteristics is used.

Generally, threshold-based crack detection approaches can achieve fast and accurate detection on ideal images that contain cracks with smooth intensity of the surroundings. However, this is impractical for real-world scenarios where there could be various noisy pixels due to background textures and lighting conditions during the data acquisition process. It is difficult to determine the optimal thresholds to distinguish crack from non-crack pixels in such scenarios.

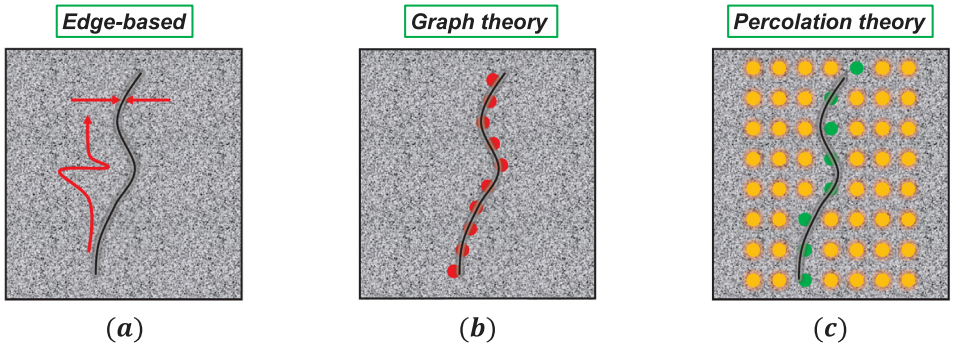

Fig. 6. An overview of morphology-based crack detection methods.

4.2. Morphology

The threshold-based algorithms detect cracks relying on only the pixel intensity information of crack images. They often failed to produce satisfying results for relatively complex scenarios such as cracks with complex topology and morphology, especially when the images suffer from poor illumination conditions. On the other hand, cracks show various morphology characteristics with regards to size, topology, continuity, contour length, etc. As such, many works use the morphology information of cracks to develop more robust crack detection algorithms. The morphological approaches are elegant and work in a manner similar to humans. In this section, we categorized existing methods into the following three categories based on how they incorporate topology/morphology information, i.e., edge-based, graph theory, and percolation theory, as shown in Fig. 6.

4.2.1. Edge-based detection algorithms

An edge in an image shows a significant local change in the image intensity, usually associated with a discontinuity in either the image intensity or the first derivative of the image intensity. Edges are often formed as a set of contiguous pixels to characterize a boundary between two disjoint regions. Intuitively, cracks are similar to edges in terms of topology and morphology. As such, many existing works have adopted edge detection techniques to achieve crack detection, which can be divided into three categories, i.e., filter methods, mathematical morphology methods, and active contours models.

Filter methods. Plenty of simple but efficient filter-based edge detectors have been developed for crack detection in the past two decades. The edge detectors are typically implemented as kernel or convolutions operators. Generally, the filter methods can be grouped into two categories, i.e., spatial domain filters such as Canny and Sobel filters and frequency domain filters such as Wavelet and Gabor filters.

The Canny detector (Canny, 1986) is an edge detection operator that uses a multi-stage algorithm to detect a wide range of edges in images. It aims to satisfy three main criteria when extracting edges of an input image, i.e., low error rate, good localization, and minimal response. The Canny algorithm has been frequently used for crack detection, e.g., by Nisanth and Mathew (2014), Abdel-Qader et al. (2003), Xu and Zhang (2013) and Cubero and Rodriguez (2017). The Sobel filter performs a 2-D spatial gradient measurement on an image and emphasizes regions of high spatial frequency that correspond to edges. Typically it is used to find the approximate absolute gradient magnitude at each point in an image. Specifically, the Sobel filters detect gradient magnitude $\sqrt{Sx^2 + Sy^2}$ and direction $\operatorname{atan}\left(C_y/C_x\right)$ of the target objects (Pereira and Pereira, 2015; Talab et al., 2016; Nisanth and Mathew, 2014; Xu and Zhang, 2013). Abdel-Qader et al. (2003) compared the Canny, Sobel, FFT, and Haar edge detection methods for crack detection on bridges surface and found that Haar transform outperforms other methods in their experiments.

However, the spatial domain filters mentioned above are typically computation and memory-consuming. They are sensitive to simple spatial transformations, e.g., translations and rotations. It has been shown that those filters often lead to false crack edge detection, e.g., due to the existence of lane marking or irregularities in the pavement surface for pavement crack detection. To overcome such limitations, many works employed frequency domain filters, e.g., Fast Fourier Transform (FFT), Fast Harr Transform (FHT), Gabor, wavelet transform, Empirical Mode Decomposition (EMD), hough transform, beamlet transforms, shearlet radon transform, Phase Symmetry-based Crack Enhancement Filter (PSCEF), etc.

The work of Abdel-Qader et al. (2003) applied FFT techniques for crack detection in concrete bridge images. Specifically, it maps an image from the spatial domain into the frequency domain to provide a totally different (but mathematically equivalent) representation of the image. They compared the effectiveness of FFT and FHT and found that the FHT was more reliable than FFT in their experiments. Gabor functions are often used for oriented and band-pass filtering, as they are more frequency-selective compared to Gaussian derivatives. In fact, Gabor filters are selective for both scale and orientation. Many existing works such as Salman et al. (2013) and Zalama et al. (2014) have employed Gabor filters to detect cracks in images of pavements and bridges.

Wavelet transform techniques have been explored extensively for defect detection, isolation, and severity evaluation in the past two decades. For example, Ouma and Hahn (2016) used 2D Discrete Wavelet Transform (DWT) to detect multi-directional and multi-scale cracks in asphalt pavements. DWT is used for low-resolution (approximate) image derivation, by thresholding the DWT approximate image at a suitable level.

Although wavelet-based methods achieved high computation efficiency, they failed to maintain the crack continuity. The decomposition of original data into the frequency domain adversely impacts the spatial entirety of cracks, thus leading to poor curvatures and continuity of cracks (Ying and Salari, 2010; Lin and Zhao, 2017). To address this issue, several works proposed combined filters for crack detection, i.e., multiple sequential filters (Nishikawa et al., 2012), matched filters (Zhang et al., 2013), particle filter (Pereira and Pereira, 2015; Lins and Givigi, 2016). The multiple sequential filters method is developed by Nishikawa et al. (2012) for crack detection on concrete material. They developed a System for the Automatic Construction of Image Filters (SACIF) based on genetic programming. Specifically, SACIF can automatically set and search the most compatible combination of filters from several training images input. Nguyen et al. (2014) assumed that cracks could be defined as objects with dark pixels and tree-like topology. They developed a Phase Symmetry-based Crack Enhancement Filter (PSCEF) to detect concrete cracks on various concrete image data. Zhang et al. (2013) proposed a matched filtering algorithm for detecting cracks on pavement images. The idea of matched filtering algorithm was proposed by O'Gorman and Nickerson (1988) to represent the characteristics of blood vessels by employing the shape,

orientation, or intensity information. Compared with five traditional edge detectors (Roberts, Prewitt, Sobel, Laplacian of Gaussian, and Canny), their method showed distinctive advantages for extracting a single crack.

The work in Pereira and Pereira (2015) and Lins and Givigi (2016) both use particle filter for crack detection. The particle filter seeks to relate the probability of an image segment to be characterized or not with a crack, based on pixel intensity and the number of pixels in its neighborhood. The key idea of particle filtering is to approximate the probability distribution using a weighted sample set. The work of Ayenu-Prah and Attoh-Okine (2008) addressed the pavement crack detection problem using Bidimensional Empirical Mode Decomposition (BEMD) combined with the Sobel edge detector, where BEMD is used to remove most noise and Sobel filter identifies cracks. In practice, pavement surface contains a lot of noise which highly influences the detection result. BEMD can decompose an image into two-dimensional Intrinsic Mode Functions (IMFs); the high-frequency instantaneous noise is removed while the remaining IMF components are used to reconstruct the final result. Moreover, BEMD is non-parametric and does not depend on a priory set.

Apart from the above classical descriptors, many sophisticated feature descriptors have also been developed, which demonstrated stronger capability to capture critical information of cracks, e.g., Local Binary Pattern (LBP) (Hu and Zhao, 2010), spatial distribution feature (Li et al., 2011), Histograms of Oriented Gradients (HOG) (Kapela et al., 2015), Crack Width Transform (CWT) (Cho et al., 2018), V-shaped features (Sato et al., 2018). It has been reported that these novel features contribute to improving detection accuracy. In general, the filter-based edge detection methods can take advantage of local patterns but neglect global information. Also, similar to threshold-based algorithms, filter-based crack detection requires significant manual efforts to determine the optimal size and format of the kernels (structuring elements).

Mathematical morphology methods. Mathematical operations such as erosion, dilation, opening, and closing, have been widely used for crack detection, which uses a structuring element for probing and expanding the shapes contained in the input image. Typically, the morphological operations are applied to binary or gray-scale images. They provide a wide range of operation tools to modify or feature the shape of objects. The morphology operations adjust cracks in the mathematical perspective. For example, binary dilation has been used to improve the connectivity of obtained cracks by changing the classification (labels) of border pixels from non-cracks to cracks.

Iyer and Sinha (2005, 2006) proposed a three-step approach to detect cracks from underground pipeline images, where morphology operations (i.e., morphological closing with a linear structuring element) have been used to enhance cracks with respect to their spatial properties. The work achieves less false detection than the Canny and OTSU methods. The work in Oh et al. (2009) employed a dilation operation to improve the connectivity between candidates' crack segments. However, the traditional binary dilation often excessively increases the width of the cracks, decreasing the detection precision. To address this concern, the work of Ai et al. (2018) extends the binary dilation operation to a weighted-dilation operation based on specific situations accordingly, which increases the crack continuity while maintaining high accuracy. In addition, Landstrom and Thurley (2012) applied mathematical operation to segment the 3D surface profile data as a first step for detecting cracks on steel slab surfaces. They first segment the data using mathematical morphology (such as opening and closing), and the resulting connected regions are assigned a crack probability using a logistic regression model. The work of Peng and Liu (2018) and Weng et al. (2019) also employed mathematical morphology operations as pre-processing steps to enhance the images. They applied morphology operations such as opening and closing operations to highlight some details of the cracks information (i.e., eliminate the burr, remove the details, and highlight the small cracks in the image).

Different from edge-based methods that typically use linear filters, morphological operations usually rely on non-linear filters. Non-linear smoothing also has an important property, e.g., it has less tendency to soften edges while filtering away high-frequency noise. On the other hand, although morphological methods work in an elegant way similar to humans, they face the challenges of manually determining the appropriate size and structure of the structuring element. In addition, the morphological operators rely on local information to improve the connectivity of cracks; purely using morphological methods does not produce promising results. As such, the morphological operators are typically used as a pre-processing or post-processing step of the entire detection process.

Active contours models. In addition to filter methods and morphology methods, several crack detection algorithms incorporate active contours models. Specifically, after segmenting cracks using adaptive threshold, the method of Tang and Gu (2013) uses a snake model with general gradient vector flow to refine the detection results. Similarly, Li et al. (2014) introduced a post-optimization method based on a modified C-V model. Since the conventional C-V model has lower location accuracy when extracting continuous crack edges, they incorporate the Canny operator into the C-V model. The Canny operator employs the non-maxima restrain and edge connection technology to suppress image noise. Nguyen et al. (2018a) proposed a level set model which uses B-spline functions to extract cracks. Specifically, the B-spline level function is employed to simulate the crack boundaries. The line-like shape of the crack and the local symmetry with respect to the crack center line is used to formulate the level-set-based mathematical model of the crack. Li et al. (2017b) developed a modified region-based active contours model combined with a Canny operator for the concrete crack feature extraction. Since the conventional active contours model used for concrete crack segmentation has a low speed and accuracy, they adjusted the level set models on the initialization using the Canny operator. Also, the work in Li et al. (2017a) combined a region-based active contours model and a level set evolution method to refine the coarse crack region, where the coarse crack region is obtained by a crack saliency map based on the OTSU threshold method. In general, active contours models typically assume a constant width of the crack, which is not practical in real-world images. Furthermore, these algorithms are difficult to be generalized and are prone to inferior results in poor imaging conditions. Also, it is challenging for them to handle slight crack edges, leading to either false-negative cases or disjoint fragments.

4.2.2. Graph theory based methods

Intuitively, cracks usually have complex topology and can be modeled as connected graphs consisting of individual crack pixels. Based on this observation, many algorithms employed graph theory techniques for crack detection over the past two decades, e.g., Minimal Path Selection (MPS), Minimum Spanning Tree (MST), and Marked Point Process (MPP), etc.

The MPS method assumes that pixels of the cracks form continuous paths of arbitrary shape. Thus, minimum paths are searched to create connected cracks. Previously, Yu et al. (2007) developed an inspection system for detecting cracks on concrete images of a tunnel. To deal with poor illumination conditions, they used histogram equalization as a pre-processing step to enhance the contrast between crack and non-crack regions. After that, the Sobel and Laplacian operators were employed to obtain the preliminary results. Then, the Dijkstra method was used to find the shortest path producing a connected crack of the graph structure. Tsai et al. (2013) presented an approach for automatic detection of the transverse and diagonal cracks using a geodesic minimal path selection model. The work of Amhaz et al. (2016) and Amhaz et al. (2014) developed crack detection algorithms for 2D images of pavements. MPS method is utilized as a pre-processing technique to select a set of minimal paths. Then, the skeleton refinement procedure, width detection technique, and Dijkstra algorithm were used to improve the quality of the cracks. Avila et al. (2014) proposed a method

for crack detection based on the MPS method that takes into account intensity and crack geometry features simultaneously. Recently, Payab et al. (2019) investigated the problem of detecting and measuring artificial cracks on concrete structures using the connected components and graph theory. Since MPS usually suffers from low computation efficiency, the work of Kaddah et al. (2019) proposed several optimization techniques to speed up the original MPS for automatic and unsupervised pavement crack segmentation.

While the above methods rely on shortest paths to connect crack fragments, some other methods construct MST to connect all crack fragments (Fernandes and Ciobanu, 2014). In the existing methods, MST is usually combined with tensor voting for crack detection. For instance, Sollazzo et al. (2016) proposed a crack detection algorithm combined with matched filtering, tensor voting, and MST method. Specifically, tensor voting was used to identify the main directions of the cracks, and MST was used to identify the crack paths. Zou et al. (2012) built a crack probability map using tensor voting and conducted a recursive tree-edge pruning process to identify desirable cracks using an MST algorithm.

The MPP method has been widely used in many application scenarios, such as pavement extraction, tree detection, wrinkle detection, etc. For example, Vandoni et al. (2016) adopted an extended MPP model to detect the crack on images of road surfaces. Their method is able to cope with noisy surfaces while at the same time avoiding false detection specific to min-cost path-based methods. However, the method requires setting the initial seed point for path walking in advance.

The graph theory-based algorithms have been demonstrated to be able to reduce false positives to some extent but suffered from a weak performance in the case of cracks with complex topologies. In addition, these methods require manually initialized positions to run the MPS/MST algorithm, which hinders real-time automatic detection.

4.2.3. Percolation theory based methods

The percolation model is a classical physical model that is widely used to characterize the phenomenon of slow escape of liquid or gas. Similar to the core idea of graph theory-based methods, percolation theory-based methods for crack detection aim to connect a series of individual pixels to form continuous cracks. Percolation is a typical local model that enhances the continuity of cracks using the intensity and shape features of the local neighboring pixels. Specifically, a percolation method starts at an initial point, then extends to the surroundings according to probability p, which is a measure of whether to percolate to the nearest neighboring region (usually calculated based on the values of local pixels).

Yamaguchi and Hashimoto (2010), Yamaguchi et al. (2008) proposed several high-speed algorithms based on percolation theory for detecting cracks on concrete surface images. The percolation theory detects the pixels in a crack and the connectivity among the pixels. It adopts the circularity feature of cracks as the terminate principle of the percolation process. Zhu et al. (2011) developed a fast percolation-based crack detection method based on the work of Yamaguchi and Hashimoto (2010). They obtained the crack geometry like width, length, and orientation automatically. Instead of performing a percolation process at each pixel of a structural image, the percolation is only calculated at the pixels with high gradient magnitudes, which significantly improves the computation efficiency. Qu et al. (2015) also proposed algorithms based on the percolation model for detecting cracks on concrete surfaces. They improved the time efficiency of the conventional percolation model by removing unnecessary percolation operations in the crack detection algorithm. In recent work, they combined the percolation model with genetic programming. Genetic programming integrates with classical filters such as Gaussian filters to construct a tree model. It is reported that the algorithm can produce promising results for detecting cracks on concrete surfaces while capable of handling some noises such as stains and water leakage (Qu et al., 2019). Xiao and Li (2018) developed a crack detection algorithm for metal-like material by combining adaptive Canny filter and percolation theory. They claimed that a pure Canny filter can maintain the details of the crack edges but neglects the breakpoints and unilateral detection. The experiments show that the fusion of the Canny filter and percolation theory can achieve more information on the crack edges with no breakpoint. The existing works also showed that it is time-consuming to apply the percolation theory method, even though many efforts have been made to improve the computation efficiency.

Compared with threshold-based crack detection methods, which solely rely on pixel intensity information, morphology-based methods have the extra capability of incorporating characteristics of crack topology. Thus, they can achieve higher detection accuracy on infrastructure surfaces even with some noises, i.e., stains and shadows. However, morphology-based methods still cannot obtain satisfying results on crack images with complex textures. In addition, the morphology-based methods typically require considerable human efforts to configure the detection algorithms, e.g., determining the optimal size and format of kernels.

To tackle the challenges of detecting cracks on images with complex textures, significant research efforts have been devoted to developing machine learning-based crack detection algorithms that can incorporate high-level descriptors/features. Over the past decade, various machine learning methods have been widely applied as a promising tool for crack detection tasks. The following sections discuss machine learning methods by dividing them into two categories, i.e., shallow machine learning methods and deep learning methods.

4.3. Shallow machine learning methods

Shallow Machine Learning (ML) methods reviewed in this section comprise ML methods with shallow structures, which have been widely used for crack detection before the deep learning methods became popular. This section reviews the existing shallow ML methods for crack detection by dividing them into two categories, i.e., supervised and unsupervised learning methods. Table 3 shows a summary of these works.

4.3.1. Supervised learning

This section introduces supervised machine learning algorithms reported for crack detection, including Support Vector Machine (SVM), Random Forest (RF), Decision Tree (DT), Sparse Representation (SR), Discriminant Analysis (DA), and some non-parametric methods like the K-Nearest Neighborhood (KNN).

SVM has been used for classification, regression, and outlier detection problems. SVM often obtains good results without significant tweaks and can work effectively for high-dimension data with insufficient samples. Li et al. (2017b) investigated crack detection on concrete structures and applied a linear SVM with a greedy search strategy to eliminate noises. Tabatabaei et al. (2019) combined SVM and RF classifier for detecting the cracks in concrete sleepers of the railway. There are two stages, where SVM is used in the first stage to reduce errors of some pre-processing steps (e.g., OTSU method) based on the general rules and shape features such as the mean of the gradient magnitude values of the frame. Ai et al. (2018) studied pavement crack detection and used an SVM model to estimate the probability that an input pixel belongs to a crack or non-crack. Chen et al. (2017) presented a two-stage SVM method to detect cracks on metallic surfaces. Specifically, a linear SVM classifier is applied for each patch in the first stage. If a patch is classified as a crack by the linear SVM, then an RBF SVM is applied in the second stage. A patch is regarded as a crack only if it is classified as a crack by both the linear and RBF SVM.

Some works also focused on developing methods to determine the optimal kernel parameters of SVM models. Liang et al. (2018) applied SVM to detect concrete cracks. The optimal kernel parameters of SVM models are selected through grid search and cross-validation with the

Table 3
Summary of crack detection methods using shallow machine learning[a].

Reference	Material	Method	Features	Auxiliary	Metrics
Peng et al. (2021)	Concrete	Ensemble model	Haar	R-FCN network	IoU
Rodriguezlozano and Leongarcia (2020)	Pavement	Ensemble model	–	Canny and dilation	$Pr\&Re$
Tabatabaei et al. (2019)	Concrete	SVM&RF	–	-	$F1$
Luo et al. (2019)	Concrete	Random forest	–	Finite state machine	$Acc\&Pr\&Re$
Wang et al. (2018)	Metal-like	Random forest	Wavelets	Region descriptor	$Pr\&Re\&F1$
Liang et al. (2018)	Concrete	SVM	Pixel intensity	OTSU	Acc
Fan et al. (2018b)	Concrete	Cluster	–	Bi-level threshold	$Pr\&Re\&F1$
Ai et al. (2018)	Pavement	SVM	Pixel intensity	Dilation	$Pr\&Re\&F1$&ROC-curve
Wang and Zhang (2017)	Metal-like	Ensemble mode	Haar	–	$Acc\&Re\&FPR$
Noh et al. (2017)	Concrete	Cluster	–	Specific filters	$Pr\&Re$
Li et al. (2017b)	Concrete	SVM	Multi	Active contour	$Pr\&Re\&F1$
Cubero and Rodriguez (2017)	Pavement	Decision tree	Projective integrals	Bilateral filter	Acc
Chen et al. (2017)	Metal-like	SVM	Texture	Bayesian	$FPR\&TPR$
Zhang et al. (2016a)	Pavement	Cluster	–	Interactive with data	$Pr\&Re\&FPR\&FOR$
Wang et al. (2016)	Metal-like	PCA	–	Hough transform	$Pr\&Re$&ROC-curve
Shi et al. (2016)	Pavement	RF	Statistical moments	–	$Pr\&Re\&F1\&CI$
Zhao et al. (2015)	Concrete	Cluster	–	Convex segmentation	$Pr\&Re\&F1$
Premachandra and Premachandra (2015)	Pavement	DA	Pixel intensity	Bilateral filter	FP
Zalama et al. (2014)	Pavement	Ensemble model	Gabor	–	$BA\&Re\&TNR$
Zhang et al. (2014)	Concrete	SVM&KNN	Histogram	Filter and dilation	Acc
Prasanna et al. (2014)	Concrete	Ensemble mode	Multi-scale	–	$Acc\&Pr\&Re$
Oliveira and Correia (2014)	Pavement	KNN&Cluster	–	Multi	$Pr\&Re$
Premachandra and Waruna (2013)	Pavement	DA	Pixel intensity	Bilateral filter	FP
Jahanshahi et al. (2013)	Pavement	SVM&KNN	–	3D reconstruction	$Acc\&Pr\&Re\&TNR$
Sun et al. (2012)	Pavement	SR	Profile signal	–	Acc&Similarity
Oliveira and Correia (2012)	Pavement	Cluster	Multi	Normalization	$Pr\&Re$
Cord and Chambon (2012)	Pavement	Ensemble mode	Texture	Fourier transform	ROC-curve
Gavilán et al. (2011)	Pavement	SVM	Texture	MDNMS	$Pr\&Re$
Hu et al. (2010)	Hybrid	SVM	HTF	Hough transform	ROC-curve
Oliveira and Correia (2009b)	Pavement	Bayes&KNN	Statistical moments	Normalization	$Pr\&Re$
Abdel-Qader and Pashaie-Rad (2006)	Concrete	PCA	–	Mean filters	$FN\&FP$

[a] IoU (Intersection over Union), Pr (Precision), Re (Recall), $F1$ ($F_1 score$), Acc (Accuracy), FPR (False Positive Rate), TPR (True Positive Rate), FOR (False Omission Rate), CI (Continuity Index), FP (False Positives), BA (Balanced Accuracy), TNR (True Negative Rate), FN (False Negatives). Please refer to Section 3 for the specific definitions of the above metrics.

improved geometric features as input, which enables better identification and classification of crack images. Hu et al. (2010) used a linear SVM classifier to determine whether a cell is a crack or not. Instead of directly using a transitional feature such as pixel intensity, they proposed a data feature based on the Hough transform. It is reported that the proposed HTF feature is invariant to rotation and illumination. Moreover, it verified that the method could produce acceptable results for various products even with the same parameters. While SVM methods demonstrated effectiveness in dealing with dimensional spaces, they typically suffer from huge computation overhead.

The RF and DT-based classifiers are also popular for crack detection problems. For example, the work in Shi et al. (2016) proposed a pavement crack detection framework based on RF technique to identify complex cracks, namely CrackForest. They used random structured forests with integral channel features to define crack tokens with structure information. Crack tokens can represent the crack regions of an image patch (16 × 16 pixels with labeled crack edges). This method is shown to be able to extract complex cracks with arbitrary topology (e.g., corners, curves, lines) on the road crack dataset CFD[5]. Wang et al. (2018) also applied structured random forests for panoramic crack detection on steel beams. They combined the structured random forests method with the semi-reconstruction method of anti-symmetrical bi-orthogonal wavelets to detect the edges of cracks. An RF-based crack classifier is used to determine the classification by voting the feature vector of each image. They used 32 × 32 crack image patch as input and 16 × 16 segmentation mask as output. The performance of their proposed methods is evaluated by comparing with traditional RF, Canny, and Roberts filters and a Fully Connected Network (FCN-8s).

Many works reported that ensemble algorithms tend to achieve better results than that of a basic ML model (Wang and Zhang, 2017; Zalama et al., 2014; Cord and Chambon, 2012). Recently, the work of Peng et al. (2021) employed AdaBoost classifier with Haar-like feature to detect bridge cracks. They first used a Region Proposal Network (RPN) to locate the crack position, then extracted crack pixels using Haar-like feature, finally refined the positive crack pixels using a local threshold method. Prasanna et al. (2014) proposed a classifier named STRUM, which contains three main components: a line detector, a spatially-tuned multiple feature extractor, and a classifier (SVM, RF and AdaBoost). It demonstrated that the boosting method (AdaBoost) achieves obviously better results as it can select and combine suitable classifiers automatically. For example, Rodriguezlozano and Leongarcia (2020) proposed an ensemble model and evaluated its performance in comparison with decision trees and a rule induction algorithm (that uses a divide-and-conquer iterative method to build the rules). They showed that the proposed method can mitigate the effect of misclassification on imbalanced datasets.

Compared with threshold-based and morphology-based methods, supervised machine learning can achieve better results due to considering more efficient features. However, the performance of supervised learning methods depends on the available datasets and manually designed crack features. The available datasets in the early stage are typically small, which limits the accuracy and robustness of shallow machine learning methods. Also, the utilization of manually designed crack features leads to significant labor efforts. To address the problem of insufficient data, many unsupervised learning algorithms have been proposed for crack detection over the past decade.

4.3.2. Unsupervised learning

Unsupervised learning relieves the burdens of manually annotating the labels at the pixel level. This section reviews the unsupervised learning methods utilized for crack detection over the past decade, including Principal Component Principles (PCA), clustering methods such as K-means clustering, fuzzy C-means clustering, and anisotropic clustering.

PCA is a powerful tool to reduce the dimensionality of data with interrelated variables, thus improving computational efficiency. It has been employed in several works (Wang et al., 2016; Abdel-Qader and Pashaie-Rad, 2006) to reduce the dimensionality of a crack dataset

before extracting the cracks on images of bridge or aircraft surfaces. For example, features such as pixel number, average gray level, and standard deviation of shape distance histogram are introduced to classify cracks by Zhang et al. (2014); and circularity, area, and slenderness ratio are used by Fan et al. (2018b). However, these features usually have a strong relationship, leading to substantial redundancy; thus, they can be reduced using PCA techniques to improve accuracy and speed.

Fan et al. (2018b) proposed a dam crack detection algorithm (CrackLG) based on local–global clustering. CrackLG enables a self-adaptive thresholding strategy for realizing image binarization. In their work, local clustering is used to distinguish cracks from noise and background based on 2D feature space (mean value and standard value of each candidate region), and global clustering is used for the 3D feature space (circularity, area, and slenderness ratio) to obtain the final crack regions. The work in Noh et al. (2017) employed fuzzy C-means clustering method to perform automatic crack detection on concrete surfaces. The performance of fuzzy c-means clustering is less influenced by the initial data. Fuzzy C-means clustering is used to choose the image clusters containing cracks as a pre-processing step, based on the probability of a pixel belonging to each cluster and its distance to the cluster. The selection is based on the average pixel intensity of the pixels belonging to a cluster. Zhao et al. (2015) developed an algorithm to extract small cracks from noisy concrete backgrounds based on the anisotropic clustering method. It treated crack images as parametric surfaces and grouped candidate pixels into arbitrarily shaped clusters by exploiting these surfaces' geometric properties. According to their elongated shapes, it can easily sift out the clusters that correspond to cracks.

The work in Oliveira and Correia (2014, 2012) proposed an automatic system for crack detection and characterization, following the guidelines defined by the Portuguese Distress Catalog (PDC). Six unsupervised learning methods have been developed for crack detection (three based on clustering techniques, while the others are density estimation, boundary, and reconstruction methods). It is reported the clustering method based on the Gaussian mixture model achieved the best overall performance. Then, Oliveira and Correia (2014) developed a toolbox in MATLAB environment called crackIT for pavement crack detection and characterization. The crackIT includes the unsupervised crack detection algorithms of Oliveira and Correia (2012) as well as other supervised learning crack detection algorithms in the work of Heijden et al. (2011). Recently, the work in Abdellatif et al. (2021) proposed a novel approach that combines the block-level and pixel-level model of CrackIT algorithm. It is reported that the approach is noise-tolerant while retaining the accurate edge localization of cracks.

Zhang et al. (2016a) proposed an interactive crack detection algorithm for a 3D pavement surface. The algorithm employed a clustering method to obtain binary annotations of the captured images (crack or non-crack). The experiments show that the algorithm can substantially improve detection accuracy based on users' feedback. Traditional interactive algorithms typically need the user to understand the algorithm and be proficient in adjusting the parameters. In contrast, this method only demands 1 of 3 feedbacks: (1) false-positive (the ratio of noises that exceed the acceptable level of the observer, i.e., ALO), false-negative (the ratio of undetected cracks is higher than ALO), or acceptable. Unsupervised learning algorithms aim to relieve the burdens of manually annotating the data samples. However, the existing algorithms still cannot achieve satisfying robustness.

In summary, ML methods make predictions by learning from input data to approximate prediction results. This overcomes the limitation of a strong requirement on data distribution. However, the shallow ML methods for crack detection usually relied on hand-crafted features to discriminate cracks from the background. This not only involves extra-human efforts but also affects the robustness of the resultant models. On the other hand, the advancement of data collection and annotation technologies have led to the era of big data, which provides opportunities for developing Deep Learning (DL) based methods for crack detection. The DL models benefit from their learnable hierarchical architectures, which could be trained to directly map input space to output space.

4.4. Deep learning

In recent years, end-to-end deep learning has become the de-facto approach for achieving accurate object detection. With the rapid development of deep learning neural networks for detection tasks, the performance of object detectors has been greatly improved. Convolutional Neural Networks (CNNs) have been shown to have the capability of capturing strong spatial–temporal dependency for image classification tasks. Many works employed CNN models to train a neural network for crack detection (Oullette et al., 2004; Xu et al., 2008; Saar and Talvik, 2010). However, the detection performance was not promising due to the limitation of the computing capabilities, dataset scale, and training techniques. Later deep neural networks are designed to learn the invariant high-level features in an end-to-end way instead of using hand-crafted features or hard-coded knowledge that need much domain expertise. This enables it to work on big and complex datasets with multiple non-linear layers, thus producing better performance. As illustrated in Fig. 7, the deep learning-based methods for crack detection tasks can be grouped into three categories: image classification, object localization, and semantic segmentation. Specifically, classification determines whether an image/patch/block contains cracks or not, object localization generates bounding boxes that cover cracks (i.e., positions information of cracks), and semantic segmentation generates a pixel-wise prediction of cracks in the images (i.e., identify cracks at pixel level). The earlier works focus more on object localization, while recent works pay more attention to semantic segmentation.

4.4.1. Image classification

Given an input image patch I, crack image classification determines whether the patch block contains a crack or not. That is, the classifier assigns a binary label to the image patch, where '0' represents a non-crack patch while '1' represents a crack patch. Table 4 summarizes the crack classification methods based on deep learning models.

In this direction, Zhang et al. (2017c) conducted several works on pavement crack detection and were among the earlier works for applying deep learning techniques (i.e., CrackNet based on CNN) for 3D asphalt pavement crack detection. They showed that CNN is superior to the SVM and 3D shadow modeling method (Zhang et al., 2017b). Later, the work in Fei et al. (2019) extended the model to CrackNet-V to improve the accuracy and computation efficiency. CrackNet-V used the VGG network for feature extraction and employed a new activation function 'Leaky Rectified Tanh' to train their deeper network.

Recently, Zhou and Song (2020) built a 3D laser-scanned pavement dataset (LRRD[16]) and developed a Deep Convolutional Neural Network (DCNN) for crack detection. They addressed two issues that have not been thoroughly addressed in existing research on DCNN-based crack classification. Since hyperparameters have a significant impact on the final prediction performance, they evaluated the performance of 36 DCNN architectures with varying hyperparameters (e.g., kernel sizes, network depths/widths, the mini-batch size, learning rate, etc.). They found that the architecture with medium-sized kernels achieved the best performance on their problem, in both accuracy and efficiency. Fan et al. (2019) proposed a pipeline for detecting cracks of pavement images, which contains three modules. Specifically, CNN determines whether an image contains cracks or not, the bilateral filtering module removes noises, and the adaptive thresholding module extracts the cracks. The work in Tong et al. (2018) and Nguyen et al. (2018b) both used a classical DCNN with Stochastic Gradient Descent (SGD) method to automatically identify and measure pavement crack length in batches. It reported that the training strategy combined with quadrature encoding and SGD could overcome the lack of labeled images

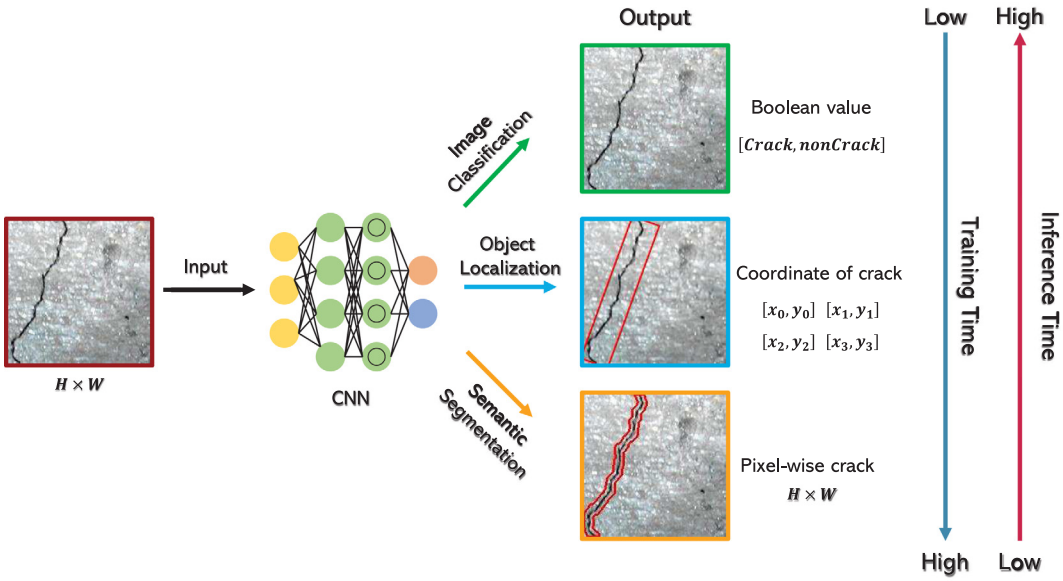

Fig. 7. An illustration of crack classification, location, and segmentation for civil infrastructures inspection.

Table 4
Summary of crack detection methods based on deep learning image classification techniques[a].

References	Material	Dataset	Base method	Transfer learning	Metrics
Wang et al. (2021b)	Concrete	299 × 299(15 000)	Inception-ResNet	√	$Pr\&Re\&F1$
Zhang and Yuen (2021)	Concrete	1024 × 1024(2068) + Concrete-Crack-Ozgene[4]	ResNet	×	$Acc\&Pr\&Re$
Zhou and Song (2020)	Pavement	LRRD[16]	–	×	$Pr\&Re\&F1$
Xu et al. (2019)	Concrete	224 × 224(1213)	–	×	$Acc\&Pr\&Re\&F1\&TNR$
Mittel and Kerber (2019)	Metal-like	1550 × 1550(1500)	AlexNet/GoogLeNet	√	$Acc\&Pr\&Re\&F1\&AUC$
Kim et al. (2019)	Concrete	227 × 227(487)	AlexNet	×	$Acc\&Pr\&Re\&F1$
Fan et al. (2019)	Pavement	Concrete-Crack-Ozgene[4]	–	×	$Acc\&Pr\&Re\&F1$
Dung et al. (2019a)	Metal-like	1024 × 768(167)	VGG16	√	$Acc\&Pr\&Re\&F1$
Tong et al. (2018)	Pavement	200 × 200(8000)	–	√	Acc
Tao et al. (2018)	Metal-like	2720 × 2040(50)	–	×	$Acc\&IoU$
Özgenel and Sorguç (2018)	Concrete	Concrete-Crack-Ozgene[4]	–	√	$AP\&F1$
Nguyen et al. (2018b)	Pavement	CrackTree[6]	–	×	$Pr\&Re\&F1$
Kim and Cho (2018)	Concrete	N/A(2073)	AlexNet	√	$Acc\&Pr\&Re$
Fan et al. (2018a)	Pavement	CFD[5]+TITS[9]	–	×	$Pr\&Re\&F1$
Doulamis et al. (2018)	Concrete	N/A(4500 + 13 500)	–	×	$Acc\&FNR\&FPR\&F1$
Dorafshan et al. (2018)	Concrete	2592 × 4608(100)	AlexNet	√	$Acc\&Re\&F1\&TNR\&PPV\&NPV$
Gopalakrishnan et al. (2017)	Hybrid	2048 × 3072(N/A)	VGG16	√	$Acc\&Pr\&Re\&F1$ &Cohen's Kappa score Blackman and Koval (2000)
Eisenbach et al. (2017)	Pavement	GAPs[11]	ASINVOS(VGG16)	×	$F1\&BER$
Chen and Jahanshahi (2017)	Metal-like	120 × 120(147k + 149k)	–	×	ROC-curve
Cha et al. (2017)	Concrete	256 × 256(40k)	–	×	Acc
Zhang et al. (2016b)	Pavement	3264 × 2448(500)	–	×	$Pr\&Re\&F1$

[a] Pr (Precision), Re (Recall), $F1$ ($F_1 score$), Acc (Accuracy), TNR (True Negative Rate), AUC (Area Under the Curve), IoU (Intersection over Union), AP (Average Precision), FNR (False Negative Rate), FPR (False Positive Rate), TNR (True Negative Rate), PPV (Positive Predictive Value), NPV (Negative Predictive Value), BER (Balanced Error Rate). Please refer to Section 3 for the specific definitions of the above metrics.

and improve the accuracy. Fan et al. (2018a) employed a CNN which takes input a small image patch of 27 × 27 pixels. They considered the imbalance problem and proposed a method that modifies the ratio of positive samples to negative samples. Their method outperformed the Canny, local threshold method, and crackForest (Shi et al., 2016) on two popular datasets CFD[5] and TITS[9]. Eisenbach et al. (2017) presented a large dataset that contains pavement distress called GAPs[11]. They employed ASINVOS net as the detection model and evaluated the effects of several techniques, including batch normalization, dropout, weight decay, and max-norm. It reported that the best generalization results were achieved by using dropout only, followed by batch normalization only, while penalizing large weights decreased the performance. Gopalakrishnan et al. (2017) used a pre-trained DCNN with transfer learning on ImageNet to detect cracks in Hot-Mix asphalt and portland cement concrete surfaced pavement. It reported that the truncated VGG-16 DCNN with 'adam' optimizer yields the best performance.

Xu et al. (2019) created a patch-based crack dataset and proposed a crack detection network based on Atrous Spatial Pyramid Pooling (ASPP) module. Their network is deeper than earlier networks and contains a total of 28 layers with an ASPP module (consisting of 10 convolutional layers) in the middle of the network. ASPP module enables the network to obtain a larger receptive field, extract multi-scale context information, and reduce the computational complexity of the model. Kim et al. (2019) presented a methodology for concrete crack detection based on CNN. In their work, crack candidate regions were

Table 5
Summary of crack detection methods based on deep learning object location techniques[a].

References	Material	Dataset	Base method	Transfer learning	Metrics
Zhou et al. (2022)	Concrete	Multi(1500)	YOLO-ED	×	mAP&$F1$
Yu et al. (2021)	Concrete	Multi(2768)	YOLO-FPM	×	mAP&PR-curve
Teng et al. (2021)	Concrete	227 × 227(990)	YOLO	√	mAP
Kalfarisi et al. (2020)	Pavement	Multi(1250)	Faster R-CNN	√	AP&DSC
Li et al. (2019a)	Concrete	N/A(3064)	Faster R-CNN	×	PR-curve&AP&mAP
Kim and Cho (2019)	Concrete	Regions(453)	Mask R-CNN	×	IoU
Mandal et al. (2018)	Pavement	416 × 416(1813)	YOLO	√	Pr&Re&$F1$&IoU
Maeda et al. (2018)	Pavement	Pavement-Crack-Maeda[13]	SSD	×	Pr&Re&IoU
Cha et al. (2018)	Hybrid	500 × 375(2366)	Faster R-CNN	×	AP&IoU
Anand et al. (2018)	Pavement	GAPs[11]+3264 × 2448(500)	SqueezeNet	×	Pr&Re&$F1$

[a] mAP (mean Average Precision), $F1$ ($F_1 score$), AP (Average Precision), DSC (Sørensen–Dice Similarity Coefficient), IoU (Intersection over Union), Pr (Precision), Re (Recall). Please refer to Section 3 for the specific definitions of the above metrics.

first obtained using a threshold method; then, they employed Speeded-up Robust Features (SURF), which is a local feature detector to efficiently extract distinctive features from the images. Özgenel and Sorguç (2018) used a pre-trained CNN on dataset Concrete-Crack-Ozgene[4] for detecting crack on concrete building images. They evaluated the detection performance regarding various impact factors, including the size of the training dataset, layers of the network, and the number of epochs for training. Kim and Cho (2018) used a pre-trained AlexNet to work on an Internet-based dataset that contains cracks, intact surfaces, and plants. Doulamis et al. (2018) combined CNN and fuzzy spectral clustering scheme for detecting cracks in tunnels. The CNN module yields only approximate regions, and then fuzzy spectral clustering refines the detected crack regions exploiting spatial and orientation coherency. The work of Dorafshan et al. (2018) compared the performance of edge-based algorithms and deep learning methods for crack detection tasks on concrete material surfaces. It was reported that deep learning methods (AlexNet) obviously outperformed edge-based algorithms (Roberts, Prewitt, Sobel, Laplacian of Gaussian, Butterworth, and Gaussian). Specifically, edge-based algorithms achieve an accuracy of 53%–79% for detecting cracks with a width of 0.1 mm while AlexNet obtains the accuracy of 99%. Cha et al. (2017) also applied CNNs to build a classifier for detecting concrete cracks. They used a CNN with 8 layers, where the input layers take patches of 256 × 256 × 3 (sampled from the original input images using a sliding window of 256 × 256). The approach achieved a detection accuracy of 98.22% on a dataset of 32K images and is able to detect cracks under various lighting conditions. However, this method produces significant errors at margin areas of cracks, and is evaluated by comparing with only some edge detection methods such as Canny and Sobel edge detectors.

Mittel and Kerber (2019) used transfer learning based on AlexNet and GoogLeNet for metal-like material crack detection, which is used for quality control. They conducted a sensitivity analysis to evaluate the effect of hyper-parameters. It reported that the multiplication factor of the last fully-connected layer has a remarkable influence on the final prediction. Tao et al. (2018) proposed a method for metal surface defect inspection, which works in a coarse-to-fine manner. In the classification stage, they used a small compact network that is suitable for metal surface defect inspection. The network is smaller than the classic classification networks such as GoogleNet and ResNet, and thus runs much faster. Chen and Jahanshahi (2017) proposed a deep learning framework called NB-CNN to detect cracks on underwater metallic surfaces from nuclear inspection videos. They used a typical CNN architecture with 4 convolutional layers, 4 pooling layers, and 2 fully-connected layers. The original dataset consists of videos with a resolution of 720 × 540 pixels. They adopted a slide window technique to extract image patches of 120 × 120 pixels in raster scan order with a step size of eight pixels. To further eliminate the false positive cases of each patch, the work in Kang et al. (2016) identifies Tubelets with Convolutional Neural Network (T-CNN) for extracting the spatiotemporal information. Then Naive Bayes was used to discarding the non-crack tubelets, where the observations of the same object in different video frames were used in conjunction with a CNN to detect objects.

The DL models discussed above typically take advantage of powerful CNNs to learn high-quality features/descriptors, and to differentiate cracks from the backgrounds. However, the detection results, as well as the algorithm efficiency, are sensitive to the setting of patch size. Thus, the above methods typically produce coarse and blocky results. Also, these approaches usually infer the existence of cracks without focusing on the morphological features of cracks, e,g., crack width and crack boundaries.

4.4.2. Object localization

Traditional crack detection methods typically rely on two techniques to detect cracks, i.e., patch extraction via sliding windows and image classification. The exhaustive sliding window scheme not only increases computation overhead but also leads to more false positives (FP) due to overlapping bounding-box detection. Also, it is difficult to determine the optimal window size due to the various resolutions of images. Furthermore, image classification methods cannot determine the exact crack locations. As such, many efforts have been made to localize cracks based on deep learning-based object localization techniques. Table 5 summarizes the crack localization methods that are based on deep learning techniques.

The crack detection methods based on object location techniques can be approximately classified into two parts: two-stage detector and one-stage detector methods. The two-stage detectors typically regard the detection problem as a classification problem, while the one-stage detectors define the detection problem as a regression problem. As a result, they have different characteristics in terms of detection performance, speed and accuracy. Typically, two-stage methods first generate candidate regions (region proposals), then use CNN to classify the candidate regions (usually also need to refine the position). They usually have relatively lower speed but higher accuracy. On the other hand, one-stage methods do not need a region proposal stage, but generate the class probability and position coordinate of the object directly. As such, they have faster detection speed but are not as accurate as two-stage methods.

Kalfarisi et al. (2020) presented a crack detection approach for detecting cracks on various infrastructures, including bridges, pavements, underground tunnels, and building inspection applications. Faster Region-CNN (R-CNN) with a structured random forest edge detection method was employed to perform cracks location and segmentation within the bounding boxes. Then, the Mask R-CNN was employed to obtain a more accurate pixel-wise mask image result. Li et al. (2019a) proposed a coarse-to-fine method to detect detection based on deep learning techniques. Specifically, a Faster R-CNN was employed to achieve coarse crack region localization and classification, and then an edge extraction step was utilized to perform fine-grained crack edge detection. They calculated different gradients along the X/Y direction to cater to cracks of different directions (e.g., transverse, longitudinal, and diagonal cracks). Kim and Cho (2019) proposed a crack detection and assessment framework for concrete structures. Specifically, a mask R-CNN is employed to conduct crack localization

Table 6
Summary of crack detection methods using deep learning semantic segmentation techniques[a].

References	Material	Dataset	Base method	Transfer learning	Metrics
Qu et al. (2022)	Pavement	Multi(5+10+3)	Attention	×	$Pr\&Re\&F1$
Guo et al. (2021)	Pavement	Multi(5+10+11+6+8)	BARNet	×	$AIU\&ODS\&OIS\&Pr\&Re\&F1$
Wang et al. (2021c)	Concrete	224 × 224	DeepLab-V3+	×	$Pr\&Re\&F1\&mIoU$
Mei et al. (2020)	Pavement	CFD5+EdmCrack60014	UNet	×	$Pr\&Re\&F1$
Ji et al. (2020)	Pavement	512 × 512(300)	DeepLab-V3+	×	$mIoU$
Huyan et al. (2020)	Pavement	N/A(3000)	CrackUNet(UNet)	×	$Acc\&Pr\&Re\&F1$
Pan et al. (2020)	Concrete	256 × 256(11 000)	SCHNet(Attention)	×	$mIoU$
Zhang et al. (2020)	Pavement	CFD5	CrackGAN(GAN)	×	$Pr\&Re\&F1\&HD-score$
Mei and Gül (2020)	Pavement	CFD5	DenseCrack(DenseNet)	√	$Pr\&Re\&F1$
Zhang et al. (2019b)	Hybrid	CFD5+TITS9+CFTD	SegNet	×	$Pr\&Re\&F1$
Zhang et al. (2019a)	Concrete	227 × 227(600)	FCN	×	$mIoU\&FWIoU$
Zhang et al. (2019c)	Pavement	4096 × 2048(6500)	CrackNet-R(RNN)	×	$Pr\&Re\&F1$
Ye et al. (2019)	Hybrid	CFD5+TITS9	CiNet(FCN)	×	$Pr\&Re\&F1\&IoU$
Yang et al. (2019)	Pavement	CFD5+CrackTree6 Crack50010+GAPs11	FPN	×	$AIU\&ODS\&OIS$
Tabernik et al. (2019)	Metal-like	KolektorSDD2	FCN	×	$AP\&(FP+FN)$
Ni et al. (2019)	Concrete	4000 × 6000(160)	FPN	√	$Pr\&Re\&F1$
Mei and Gül (2019)	Pavement	EdmCrack60014	GAN	√	$Pr\&Re\&F1$
Liu et al. (2019a)	Concrete	512 × 512(27)	UNet	×	$Pr\&Re\&F1$
Liu et al. (2019c)	Hybrid	DeepCrack3+TITS9	DeepCrack(FCN)	×	$Acc\&mIoU\&Pr\&Re\&F1\&AUC$
Liu et al. (2019b)	Pavement	CFD5	FPCNet(SegNet)	×	$Pr\&Re\&F1$
Li et al. (2019b)	Concrete	4032 × 3016(1375)	FCN	√	$mIoU\&FWIoU$
Fei et al. (2019)	Pavement	4096 × 2048(6500)	CrackNet-V(FCN)	×	$Pr\&Re\&F1$
Dung et al. (2019b)	Concrete	Concrete-Crack-Ozgene4	FCN	√	$F1\&AP$
Bang et al. (2019)	Pavement	1920 × 1080(100)	SegNet	√	$Pr\&Re\&F1\&IoU$
Zou et al. (2018)	Pavement	CrackTree6	DeepCrack(SegNet)	×	$ODS\&OIS\&AP$&PR-curve
Yang et al. (2018)	Concrete	224 × 224(800)	FCN	×	$Pr\&Re\&F1$
Kobayashi (2018)	Concrete	5472 × 3678(14)	SpiralNet(UNet)	×	$Pr\&Re\&F1\&mAP$
Zhang et al. (2017c)	Pavement	4096 × 2048(2000)	CrackNet(FCN)	×	$Pr\&Re\&F1$
Schmugge et al. (2017)	Concrete	N/A(685)	SegNet	×	$Pr\&Re\&F1$

[a] Pr (Precision), Re (Recall), $F1$ ($F_1 score$), ODS (Optimal Dataset Scale), OIS (Optimal Image Scale), $mIoU$ (mean Intersection over Union), Acc (Accuracy), $HD-score$ (Hausdorff Distance), $FWIoU$ (Frequency Weighted Intersection over Union), IoU (Intersection over Union), AP (Average Precision), FP (False Positives), FN (False Negatives), mAP (mean Average Precision). Please refer to Section 3 for the specific definitions of the above metrics.

and segmentation. Then, the crack widths were further quantified using some image processing techniques such as median filtering, blob removal, and the OTSU binarization method. The Mask R-CNN is trained on 1102 crack regions masked on 376 concrete images. It reported that the algorithm could detect most of the cracks with a width of 0.3 mm or wider for concrete structures. Cha et al. (2018) developed a Faster R-CNN based approach to detect multiple types of damages in civil infrastructures, such as cracks, steel corrosion, delamination, bolt corrosion, etc. They used ZF-Net (Zeiler and Fergus, 2014) as the backbone which has faster training and testing speed in comparison to networks like VGG16, ResNet152 or GoogleNet. Moreover, to detect multiple types of damage, ZF-Net was modified by replacing the last max-pooling and fully connected layers with the sliding convolutional kernels. Also, its softmax layer is replaced with regression layers. In addition, the work in Anand et al. (2018) proposed a SqueezeNet based algorithm for detecting road cracks and potholes. They removed the last convolution layer of the SqueezeNet and incorporated an encoding layer with dictionary learning and residual encoding methods.

The work in Majidifard et al. (2020) collected 7237 google street-view road images which are annotated as transverse crack, alligator crack, longitudinal crack, etc. They developed an asphalt pavement condition index system relying on You Only Look Once (YOLO) and UNet, where the YOLO locates the cracks and the UNet quantifies their severity simultaneously. Mandal et al. (2018) developed an automated pavement distress analysis system based on the YOLOv2 deep learning framework. The detection model is pre-trained on the MS-COCO dataset (Lin et al., 2014) and then tuned on the crack dataset. It reported that the model is accurate in detecting alligator cracks but struggles with transverse cracks. Maeda et al. (2018) developed a large-scale road damage dataset, Pavement-Crack-Maeda[13]. They proposed a crack detection algorithm based on a Single Shot MultiBox Detector (SSD) network. The SSD that uses Inception-V2 and MobileNet as backbones were tested on the dataset and achieved recall and precision larger than 77% and 71%, respectively.

The above-discussed methods typically could catch up with R-CNN in detection accuracy but ran much faster, which is critical for civil infrastructure inspection. The acceleration is obtained by adopting selective search techniques such as replacing traditional sliding windows or bounding boxes. Cracks usually have diverse morphologies with varying sizes, shapes, widths, lengths, and orientations. The existing works typically used a fixed size of the bounding box (height, width, as well as their ratio), which cannot properly capture the cracks. In addition, most methods use a general object detection framework, which does not distinguish between cracks and general objects. And the resultant cracks may not be suitable for further quantitative assessment of the cracks.

4.4.3. Semantic segmentation

The classification and localization methods discussed in the previous sections cannot achieve crack detection at the pixel level. For example, patch-based methods typically predict the existence of cracks while giving insufficient emphasis to crack shapes and boundaries. The detection accuracy is insufficient for assessing the physical and functional conditions of the civil infrastructures. Many engineering problems such as behavior analysis of crack evolution process demand accurate detection at pixel-level, i.e., determining the presence and the exact boundaries of cracks. To overcome such limitations, a large number of semantic segmentation networks have been developed to segment cracks at the pixel level without requiring any prior knowledge. Table 6 summarizes the semantic segmentation methods for crack detection using deep learning techniques.

FCN-based networks. The Fully Convolutional Networks (FCN) was developed based on the original CNN structure by Long et al. (2015) in 2014. Compared with image block classification methods, FCN structure allows a varying size of input images and significantly accelerates the processing speed. Thus, lots of subsequent methods have been proposed based on FCN to obtain semantic segmentation of cracks.

Ye et al. (2019) proposed a Ci-Net for structural crack identification based on FCN. Compared with FCN, Ci-Net has the same architecture but different convolution and deconvolution layers. However, they only evaluated the performance by comparing Ci-Net with edge detection methods on two datasets (CFD[5] and TITS[9]). Liu et al. (2019c) proposed a network deepCrack based on FCN and Deeply-Supervised Nets (DSN) to detect cracks on a public concrete/pavement crack dataset (DeepCrack[3]). Different from FCN, which only uses the last convolutional layer, they made many modifications to adapt to their problems, including (1) A DSN module to provide integrated direct supervision for multi-level features learning; (2) a cross-entropy loss function which combines the final fused prediction and side-output layer prediction; (3) incorporating the Conditional Random Field (CRF) and Guided Filtering (GF) methods to refine the fused prediction. Li et al. (2019b) proposed an FCN-based multiple damages detection method for concrete structure. It reported that their methods not only effectively identify cracks but can detect spalling, efflorescence, hole, and their combinations. They show that their FCN-based approach obtained slightly better performance than SegNet.

Dung et al. (2019b) proposed an algorithm to detect cracks on concrete material based on FCN. Their work selected different CNN architectures as encoders to extract features, e.g., VGG16, Inception-V3, and ResNet. Tested on a public dataset Concrete-Crack-Ozgene[4], it reported that VGG16 outperformed Inception-V3 and ResNet for crack image classification. Yang et al. (2018) also used FCN to extract cracks on buildings and infrastructures. It reported that although the proposed FCN-based algorithm produces lower accuracy than CrackNet, it is much faster and works in an end-to-end model that requires no post-processing or pre-processing. Also, a crack skeleton method was proposed to quantitatively measure the morphological features of cracks. Although FCN based methods demonstrated obvious improvements for semantic segmentation of pixel-level crack detection, they failed to consider local and/or global context-dependency among crack pixels. In other words, there is still much space for potential performance improvement.

In FCN, pooling is used to increase the receptive field to reduce the image size, while up-sampling restores the image size. However, this process causes a loss of accuracy, so dilated convolution is proposed to reduce this loss. Liu et al. (2019b) proposed a Fast Pavement Crack Network (FPCNet) using an encoder–decoder structure. Specifically, FPCNet consists of two modules, a Multi-Dilation (MD) and a SE-Upsampling (SEU) module. MD extracts the crack features of multiple several context sizes via dilated convolution with multiple rates. SEU restores the resolution of the MD features with extra operations. Zhang et al. (2019a) also adopted FCN with dilated convolution for concrete crack detection. In their work, a residual network and dilated convolutions with various dilation rates were used to extract the feature maps of diverse receptive fields. Their proposed model outperformed some classical FCNs such as FCN-VGG and FCN-ResNet-5/18. Similarly, in optimizing pooling operation, the work in Ji et al. (2020) proposed a neural network for pavement crack detection based on DeepLab-V3+. DeepLab-V3+ also used dilated (atrous) convolution to capture multi-scale information by controlling the resolution of features and adjusting the filter's field-of-view. In the decoder module, the operation of bi-linear up-sampling is employed to decode the feature map.

UNet-based networks. Arguably, UNet-based architectures are more popular and successful for crack or damage detection tasks (Liu et al., 2019a; Ronneberger et al., 2015). The advantage of UNet is the introduction of skip connection, which fuses the feature maps of the encoding stage with the feature maps of the decoding stage, helping to the recovery of segmentation details (e.g., edges and contours).

Huyan et al. (2020) proposed a deep learning model based on the UNet, called CrackUnet, for pixel-wise pavement crack detection. CrackUnet differs from the original UNet in that: (1) it does not require pre-trained network models or data argumentation; (2) fixed convolutional kernel size is used to extract multi-level features for simplicity; (3) it uses a max-pooling layer instead of a mean-pooling layer since the pavement cracks and background typically have high contrast. The work in Ai et al. (2020) developed a variant of the UNet model to detect cracks on rock surfaces. They used a new way to implement skip connections, which allows their network to merge multi-level features to obtain intermediate results that can be better combined with subsequent processing approaches for achieving additional improvement. By exploiting the UNet and the connectivity of pixels of pavement cracks, Mei et al. (2020) proposed a deep semantic network with an effective loss function to extract pavement cracks. The work analyzed the drawback of the regular cross-entropy loss function for pixel-level crack annotations and proposed a new loss function considering the connectivity of pixels. They claimed that the loss function is a sum of the cross-entropy function of all eight connectivity maps. It reported that the proposed method could increase the F1 score from 70.95% to 75.35% on the EdmCrack dataset[14].

SegNet-based networks. Similar to FCN, SegNet is also an encoder–decoder structure, which was proposed by Cambridge to solve image semantic segmentation in automatic driving tasks (Badrinarayanan et al., 2017). The innovation of SegNet is the up-sampling processing method of decoder for extracting low feature maps. Specifically, the decoder uses the pooling indices calculated in the max-pooling process to calculate the non-linear up-sampling of the corresponding encoder, which eliminates the process of upsampling learning.

The SegNet (Schmugge et al., 2017) directly employed is to get preliminary results in nuclear power plant inspection. Then, overlapping frames and pixel-level probability maps obtained from SegNet are fused to improve the accuracy. Rather than classifying each pixel using the information of a single frame, they fused features from multiple frames based on the BoofCV mosaicing algorithm (Abeles, 2013). A DeepCrack (Zou et al., 2018) is developed based on SegNet architecture for pavement crack segmentation. Their method pairwisely fuses the convolutional features generated in the encoder network with those generated at the same scale in the decoder network. The proposed method outperformed some edge detection algorithms like HED and RCF. Instead of using VGG-16 in SegNet, the work of Bang et al. (2019) using ResNet-(50, 101, 152, 152a, and 200) as encoder backbone. They trained and tested the models on a pavement infrastructure dataset with 427 training and 100 testing images with a resolution of 2560×1440, and reported that the network based on ResNet-152 achieved the best scores. In addition to the above variants, a deep semantic segmentation network (Zhang et al., 2019b) is proposed with a context-aware fusion algorithm. This method integrates the pixel-wise prediction results from multiple local overlapping image patches. In addition, this method also supports input images of arbitrary sizes.

FPN-based networks. Instead of simply adding features of different scales in FCN, Feature Pyramid Networks (FPN) uses 1×1 convolution operations to process shallow feature maps and then adds them with high-level features in the feature fusion stage. Recently, Yang et al. (2019) proposed a Feature Pyramid and Hierarchical Boosting Network (FPHBN) based on the Holistically-nested Edge Detection (HED) method. FPHBN applied a novel loss function with a 'weighted-fusion' layer to automatically deal with class imbalance, i.e., few positive pixels (crack) and more negative pixels (non-crack). Their model is able to learn robust feature representation to cope with the crack detection problems with highly skewed classes. Ni et al. (2019) proposed an automatic FPN-based crack delineation framework for structural inspection, called Crack Delineation Network (CDN). They used GoogLeNet as an encoder model and employed FPN to achieve feature-map fusion based on the output of GoogLeNet.

Other networks. In general, the recurrent neural network (RNN) is first proposed to solve the natural language processing (NLP) tasks. Based on the assumption that cracks can be considered as a sequence of pixels, the work in Zhang et al. (2019c) proposed a network CrackNet-R based

on Gated Recurrent Multi-layer Perceptron (GRMLP) to achieve more robust detection results. Specifically, CrackNet-R consists of two parts: (1) sequence generation finds the best local paths of cracks, and (2) sequence modeling predicts timely probabilities of the input sequence being a crack pattern. Compared with the original CrackNet of Zhang et al. (2017c), their method achieves improvements in both detection speed and accuracy.

Generative Adversarial Network (GAN) (Goodfellow et al., 2014) can learn to generate new data with the same statistics as the training set. This can help solve the problem of limited positive crack data in crack detection, regarding the crack detection task as an image-to-image translation problem. The work in Pei et al. (2021) and Mei and Gül (2019) both proposed a pavement crack detection algorithm using conditional Wasserstein Generative Adversarial Network (cWGAN) and deep convolutional generative adversarial networks (DCGANs). Zhang et al. (2020) proposed a CrackGAN network for crack detection on pavement images which used UNet in the generator part and a pre-trained discriminator using DCGAN with label crack patches in the discriminator part. The FCN-based crack detection methods often face the "All Black" issue, that is, most of the pixels in an image are background, and the network may easily converge to the status that treats all the pixels as background and still achieves a very small loss value. The work reported that their methods could well tackle the "All Black" issue, and the CrackGAN achieves the best performance on four crack datasets in comparison to the baseline methods.

The work in Mundt et al. (2019) used meta-learning approaches to find suitable CNN architecture for concrete defect detection. They published a large-scale dataset, called COncrete DEfect BRidge IMage dataset (CODEBRIM),[17] which features six mutually non-exclusive classes: crack, spallation, efflorescence, exposed bars, corrosion (stains), and non-defective background. They showed that meta-learned methods achieve equivalent or better accuracies while being more parameter efficient. A CNN-based architecture (Tabernik et al., 2019) is proposed for the detection and segmentation of metal-like surface anomalies. In their public dataset KolektorSDD[2], crack is the main type of anomaly. Their model consists of two parts, i.e., a segmentation network and a decision network. The first part tries to increase the receptive field size as meta-like surfaces usually contain tiny cracks, while the second part tries to avoid using too many feature maps in the decode stage. A spatial-channel hierarchical network called SCHNet (Pan et al., 2020) is proposed for crack segmentation on the concrete bridge deck. They first used VGG19 to extract the feature and employed a self-attention mechanism to capture long-range contextual information in both the spatial and channel dimensions. They tested their work in a public dataset SDNET2018[12] (selected 11000 crack and non-crack images of the bridge deck) and reported that the SCHNet outperforms U-net, DeepLab-v2, PSPNet, Ding, and Dilated FCN in pixel-level results. The work in Guo et al. (2021) proposed an encoder–decoder network, named BARNet, to detect road cracks. Different from the above methods, the BARNet consists of three parts, i.e., a Base Predictor Module, an Edge Adaptation Module, and a Refinement Module. The BARNet can overcome the problem of inaccurate boundaries for crack localization. Their results show improvements on five popular public datasets, i.e., CFD[5], Crack500[10], GAPS384[11], CrackTree[6], and Pavement-Crack-Chambon[8].

Very recently, Liu et al. (2021) proposed a Crack Transformer network, called CrackFormer, for crack detection. CrackFormer is mainly composed of SegNet with attention modules and has an advantage for detecting fine-grained cracks. The results show that CrackFormer has achieved a higher pixel-level crack detection accuracy than some SOTA fine-grained crack detection methods, like DeepCrack (Zou et al., 2018).

The above-discussed semantic segmentation methods for pixel-wise crack detection typically achieved better detection accuracy and speed in comparison to image classification and localization methods. However, the existing methods still face the challenge of imbalanced data, i.e., crack pixels only account for a tiny proportion of many images. This could lead to lots of false negatives (i.e., classify a crack pixel as a non-crack pixel) while still generating high accuracy.

4.4.4. Transfer learning

Due to limited available data, it is challenging to train a deep neural network with high performance using a single crack dataset. To address this challenge, many works adopted transfer learning techniques to improve crack detection performance by using a pre-trained network obtained on a larger similar dataset (Pan and Yang, 2009). Transfer learning stores the model of existing tasks and uses it on other different but related problems. For example, the knowledge (or models) used to identify cars can also be used to improve the ability to identify trucks. In practice, transfer learning transfers the knowledge from one domain to another domain. As a result, the target domain can gain more satisfactory learning results. Generally, the data volume in the source domain is adequate, while the data volume in the target domain is small. It migrates the knowledge learned when the data volume is sufficient to a new environment with a small data volume, which is just the case for the crack detection task. Transfer learning has also been proved to enhance training efficiency and accuracy for crack detection tasks.

Table 4 (as introduced in Section 4.4.1) has pointed out the works that employed transfer learning techniques for crack image classification. Specifically, Mittel and Kerber (2019) employed AlexNet and GoogLeNet trained on ImageNet as a pre-trained network. Dung et al. (2019a) compared three deep learning methods, i.e., a shallow CNN, a tuned pre-trained VGG16 architecture, and a fine-tuned top layer of VGG16, for cracks detection at gusset joints in steel bridges. It reported that the fine-tuned methods achieve promising performance, indicating that robust crack classifiers can be trained efficiently on a limited crack image dataset using transfer learning techniques. Different from the above works, the work in Tong et al. (2018) transferred the weights and the bias from a CNN, which is trained on a crack dataset consisting of cracks of various widths. Özgenel and Sorguç (2018) reported that the pre-trained networks could be successfully transferred for crack detection even if they are trained on completely different datasets. They also claimed that the features learned in one material could transfer to other materials in their public dataset. Kim and Cho (2018) used a pre-trained AlexNet whose final layer is modified to five classes: crack, multiple lines, single line, intact surface, and plant. Dorafshan et al. (2018) used AlexNet-based DCNN with transfer learning for concrete crack detection. It proved that the crack detection performance is more effective if the network is previously trained on the structural defects dataset. Their work also showed that a network with pre-trained DCNN was able to detect fine cracks which were wider than 0.04 mm. Gopalakrishnan et al. (2017) used a pre-trained VGG network with transfer learning on ImageNet to achieve pavement distress detection. It reported that the network with 'adma' optimization algorithm retrained on a VGG-16 DCNN that is pre-trained on ImageNet exhibited the best results.

Table 5 (as introduced in Section 4.4.2) has pointed out the works that employed transfer learning techniques for crack localization. Kalfarisi et al. (2020) trained the Inception-V2, ResNet-101, ResNet-50, and Inception-ResNet-V2 on the MS-COCO dataset to obtain initial models, which are then fine-tuned on a bridge crack dataset that contains 1250 images with a resolution of 344×296 to 1024×796 pixels. In addition, some data augmentation operations such as random rotations, random vertical, and random horizontal flips were adopted to help avoid over-fitting. Mandal et al. (2018) also used a transfer learning technique to train the YOLOv2 network on the MS-COCO dataset for the pavement crack detection task. Specifically, they froze the weights of ResNet-101 and RPN throughout the training and fine-tuned the location-sensitive score maps and the output layer.

[17] https://zenodo.org/record/2620293.

Table 6 (as introduced in Section 4.4.3) has pointed out the works that employed transfer learning techniques for semantic segmentation. Ni et al. (2019) used a pre-trained GoogLeNet model learned from the ImageNet for structural crack detection. After obtaining the pre-trained GoogLeNet, they used 64k crack patches and 64k non-crack patches with 224 × 224 pixels to fine-tune their model. Mei and Gül (2019) proposed a generator for detecting pavement cracks, which was first pre-trained on ImageNet, and then fine-tuned on a small crack dataset, i.e., CFD[5]. Li et al. (2019b) used transfer learning to initialize the parameters of an FCN in their work. Their model was first built on a DenseNet-121 network and then retrained on their own dataset to obtain a model for detecting damages on concrete material, including cracks, spalling, efflorescence, and holes. They reported that the pre-trained parameters of the DenseNet-121 would significantly decrease the entire training time of FCN. Dung et al. (2019b) adopted three pre-trained CNN models, i.e., VGG16, Inception-V3, and ResNet, to their work. It reported that the VGG16 and Inception-V3-based classifiers achieved promising performance for their problem. Bang et al. (2019) used the VGG-16 and ResNet-50(101, 152, 200) as an encoder with transfer learning based on ImageNet datasets to detect road cracks in their work. It reported that ResNet-152 with transfer learning yields the best performance.

To sum up, deep learning-based crack detection methods have achieved significant improvements in both accuracy and speed. However, the existing deep learning models are typically trained to adapt to the characteristics of a specific dataset of crack images. Due to the inherent differences in crack topologies, noisy surroundings, and illumination conditions, one model that is customized for a specific detection task (dataset) often performs poorly on another detection task (dataset). Also, compared with other methods, deep learning models are much more complex, with up to thousands of million parameters. Thus, training a successfully deep neural network requires a considerable amount of data as well as powerful hardware computation platforms, which could be impossible/challenging for some application scenarios. For instance, the work in Özgenel and Sorguç (2018) has trained a large deep model with ResNet-152 backbone using nearly 40k RGB images with concrete cracks and achieved nearly a 0.998 F1-Score. However, it failed to reproduce satisfying results in an asphalt pavement application.

5. Discussion

Although researchers have proposed many computer vision-based methods to tackle crack detection tasks over the past 20 years, this domain has not yet been fully explored, and several challenges remain unresolved. This section discusses some of the main challenges and provides insights on future prospects.

5.1. Current limitations and challenges

5.1.1. Limited datasets

The lack of publicly available datasets with proper annotations is one of the critical barriers that hinder the development of this field, due to the uneven distribution of crack data and the need for time-consuming and laborious intensive work for data preparation. In fact, it is impractical to build large datasets by directly merging multiple data sources of cracks. This is because different works target cracks of various types and granularities, and the ground truth datasets constructed by existing studies are labeled at different levels (e.g., pixel level, block-level, or image level).

It is imperative to establish standard public benchmark datasets for evaluating the effectiveness of crack detection approaches. Ideally, a standard benchmark dataset should (1) be large-scale with respect to the total number of crack images of different materials and cater to various application scenarios; (2) contain cracks of various categories and morphologies, and has a large range of size variations, with the cracks having high inter-class similarity and intra-class diversity; (3) consist of sufficient image variations due to different imaging conditions (e.g., with multiple lighting conditions using the illumination sources placed at multiple heights), weather, seasons, and image quality. In addition, it is desirable to have high-quality (ideally manual) annotations with careful inspection in each frame to enable researchers to develop and validate their methods qualitatively and quantitatively. The dataset should also be carefully tailored for methods targeting multiple levels of detection granularity (e.g., image level, block level, pixel level, or polygon style labels).

5.1.2. Imbalanced classes (crack and non-crack pixels)

Real-world data usually show long-tailed distributions with severe class imbalance, posing significant challenges for detection models. In the crack detection problem, a crack image has much more background (non-crack) pixels than crack pixels, leading to an imbalanced classification problem. The class imbalance in the dataset(s) can dramatically skew the performance of classifiers, introducing a prediction bias in favor of the majority class. As a result, it becomes rather tricky to discriminate between the non-crack and crack pixels effectively. Such a biased learning process could classify all pixels as non-crack pixels and produce a deceptively high accuracy metric.

In addition, the crack images usually contain a large number of noisy pixels, which have similar intensity and texture information as the cracks, making it difficult to detect cracks at pixel level. Although some efforts have been made to detect objects on imbalanced data with the presence of noises, their proposed strategies are not effective on heavily imbalanced data such as crack images. For example, cracks typically represent less than 1% of the pixels from road images. Such a rate of imbalance poses a great challenge for effectively training a model.

Such challenge is compounded by the existence of inaccurate and unreliable pixel-level labels at the boundaries of cracks. Such unreliable labels are special cases of noisy pixels. To mitigate the impact of such noisy labels, many works adopted tolerance margins for evaluation. Unfortunately, tolerance margins are lax and tolerant to errors at the pixel level, providing too optimistic scores. Although some state-of-the-art DL-based methods achieved an accuracy of 95% with the assumption of tolerance margins, it is still insufficient for calculating crack quantification index (crack severity) and further evaluating the damage degree of infrastructures. It is challenging to increase the detection accuracy without using tolerance margins considering that cracks account for only a tiny portion of the images.

5.1.3. High inference time and sophisticated network

DL-based crack detection methods were shown to achieve better performance than traditional methods due to the powerful learning ability of neural networks, but at the cost of more computation, longer inference time, and complex parameter setting. In the past decade, DL-based detectors have become more and more sophisticated. However, most of the existing DL-based methods rely on the manual design of deep neural networks and the empirical setting of hyper-parameters. As such, the existing model structures typically are not optimal solutions for crack detection. Reducing human intervention when designing a powerful crack detection model is a desirable but challenging task. On the other hand, many detection algorithms need to be deployed on embedded computing systems mounted on mobile platforms such as UAVs in order to achieve real-time detection with low delays and communication overheads. However, little efforts have been made to develop crack detection algorithms that can run efficiently on edge or low-power devices.

5.2. Potential future directions

5.2.1. Semi-/weakly-supervised learning

Semi-/weakly-supervised learning provides the means to pre-train a neural network using unlabeled data, followed by fine-tuning for subsequent tasks with limited annotations. Various semi-supervised methods have been proposed in the past decade to address the challenge of insufficient data for salient object detection as well as semantic segmentation (Zhang et al., 2021; Wang and Su, 2021), such as few-shot learning (Mehra and Hamm, 2021), attention mechanism (Zhou et al., 2021) and contrastive learning (Wang et al., 2021a). These methods achieve domain adaptation by sharing knowledge from a labeled original domain with the target domain without labels. Although significant progress has been made in semantic segmentation using supervised methods, it remains challenging to deploy the segmentation models in unseen domains due to domain biases.

Unlike conventional methods that train the DCNNs with a large number of labeled data (Jiaxu et al., 2021), few-shot learning has recently attracted significant attention to adapt DCNNs to new classes with few annotated samples. Many works have explored the use of implicit cues in additional datasets without target-domain supervision to support few-shot detectors to enhance robust task notions (Lin et al., 2021). For instance, the work in Rill-García et al. (2022) used a label noise correction technique based on weakly supervised learning. They introduced a novel synthetic dataset (Syncrack) and tested label correction methods. The results show that their method performs better in datasets with inaccurate annotations. The few-shot object detection methods have demonstrated promising potential to reduce costs for data collection and annotation. Therefore, it is encouraging to develop few-shot crack detection methods, which rely on as little supervision as possible, to tackle the challenges of insufficient training data. Specifically, an existing network can be used as the fundamental framework and redesigned for incremental few-shot crack detection, e.g., by introducing meta-learning techniques to adapt the model to unseen knowledge.

Recently, contrastive learning has been proposed to further augment the generalization capability of deep learning models with limited resources. Contrastive learning has demonstrated great potential in learning latent representation of images even without any label and achieved promising performance in the area of self-supervised representation learning (Chen et al., 2020). It builds a framework that can self-learn representations, which does not suffer from the need for many labels and can learn robust and general representations. Contrastive representation learning aims to learn an embedding space in which similar sample pairs stay close to each other while dissimilar ones are far apart. Contrastive learning can be applied to both supervised and unsupervised settings. When working with unsupervised data, contrastive learning is one of the most powerful approaches in self-supervised learning. For example, contrastive learning can be used as an additional auxiliary training objective that acts as a data-dependent regularizer to promote more general and transferable features. The learned transferable embeddings help the detection model to generalize to different crack types and noise conditions.

5.2.2. Novel loss function

There exist many techniques for general object detection that has not yet been applied to address crack detection problems. For example, many efforts have been made to develop novel loss functions for object detection and semantic segmentation problems on imbalanced datasets, such as focal loss (Liu et al., 2021b) and Tversky loss (Jadon, 2020). Specifically, focal loss (Lin et al., 2017) was frequently used to mitigate strong class imbalance by focusing on harder samples and producing lower gradients on easy samples. The work in Ali et al. (2021) proposed a local weighting factor with a sensitivity map to remove the network biasness for crack pixels. An Average Localization Recall Precision (aLRP) loss for object detection was proposed by Oksuz et al. (2020), which demonstrated that a simple loss function could provide 5% performance gain on RetinaNet. Several other loss functions have also been reported and evaluated for object detection, such as weighted cross-entropy and distribution-balanced loss (Wu et al., 2020), etc.

In addition, learning discriminative image representations plays a vital role in detection problems on imbalanced datasets as better features make better detectors. As discussed before, contrastive learning achieved promising performance in representation learning; thus, it is reasonable to explore effective contrastive learning strategies (both supervised and self-supervised) and tailor them to learn better image representations from imbalanced data in order to boost the detection accuracy thereon. In particular, the contrastive loss can be combined with cross-entropy loss such that the former is used to learn image/crack, while the latter is used to train the detector.

5.2.3. Lightweight network

As discussed, deep learning-based detectors are becoming increasingly sophisticated due to the need to incorporate additional modules to improve detection robustness. Unfortunately, they not only increase the complexity of training optimal models but also the detection overhead and response delays. As such, it is necessary to design lightweight neural networks, which reduce the computational effort and enable them to run efficiently on low-cost devices in real-world applications. Knowledge distillation has been used as a method for model compression. Unlike pruning and quantization in model compression, knowledge distillation trains a small model by building a lightweight model and using supervised information from a larger model with better performance in order to achieve better performance and accuracy.

Also, the specific application of crack detection has unique characteristics compared to general computer vision tasks, which can be used to optimize network structures. Many efforts have been made to develop efficient techniques of neural architecture search (NAS) and automated machine learning (AutoML) to automatically search for the optimal model for a specific task scenario. For example, the work in Hou et al. (2022) proposed a method that uses lightweight networks based on MobileNet and MobileNet-SSD and transfer learning to inspect pavement conditions. The approach can achieve efficient distress classification and detection without requiring a large amount of fine-labeled data and computation resources.

5.2.4. Crack evolution behaviors

In practice, cracks propagate through specific structures leading to structural weakening and failure. For example, stress concentration and crack propagation could occur if the loading on a bridge structure exceeds its expected value. Cracks also considerably impact the rate of moisture and chemical ingress into concrete. This further leads to serious instability when the cracks evolve. Therefore, understanding how cracks evolve becomes increasingly important for continuous and periodic diagnosis and inspection of civil infrastructures in real-world scenarios. In particular, being aware of the changing trend of the occurrence and severity of cracks facilitates the prediction of future conditions, supports predictive maintenance, and manages limited repair resources, which ensures the safety quality and service requirements of civil infrastructures with limited maintenance effort. Thus, it is crucial to monitor the occurrence and evolution behaviors of infrastructure cracks.

Compared to other defects like potholes and joints, cracks show strong continuous structural characteristics (e.g., propagation) that are predictable. Typically, cracks propagate through a specific structure which causes weakening leading to structural failure. Existing works show that the generation of crack is interpretable from the perspective based on classical physical theory. Crack evolution has been modeled as a procedure of evolutionary geometry, which is a critical indicator reflecting the structure conditions and failure behaviors (Khasin, 2014; Bayar and Bilir, 2019; Bernstone and Heyden, 2009).

There have been few preliminary studies that monitor the crack evolution/propagation behaviors (Ai et al., 2020; Kong et al., 2021). For example, the work in Ai et al. (2020) conducted an experiment to simulate the crack evolution process. A high-speed camera captures the evolution process as video data, consisting of a sequence of images. They also proposed a method to detect evolving cracks in the obtained video data using deep learning-based models. Amin et al. (2018) considered the detection of small cracks in the preliminary stages of nucleation and their growth, which significantly reduce catastrophic failures. Crack growth testing is important to characterize the expected behavior during service. Tavares et al. (2016) developed a dedicated illumination device to assist in the full automation of fatigue crack-growth measurements. It is well known that the presence of the crack in the structure also changes the dynamic response of the structure. In other words, the crack in the structure results in changes in frequencies and mode shapes for the vibration. Therefore, the vibration-based damage identification methods have been widely studied for decades (Li et al., 2009) to model the characteristics of crack evolution (in terms of the depth, length, location, propagation velocity, etc.) based on the environmental parameters/variables (i.e., vibration wave, energy flow, and stress/strain). Recently, the work in Perry et al. (2022) used UNet and Gaussian Process surrogate models to extract cracks and predict stress intensity factors K of a steel structure. The work can empower an inspector to easily identify steel cracks, predict their propagation from a collection of raw images, and make rapid repair decisions. It is worth highlighting that the existing works studying crack evolution/propagation typically rely on measurements like strain, frequency, and vibration rather than image data. Since most state-of-the-art for crack detection focus on analyzing the existence of cracks on image data, it is necessary to investigate the crack evolution behaviors based on image data.

5.2.5. Robust automatic inspection systems

Significant efforts have been made on crack detection as reviewed in the previous sections; however, the application of automated vision-based crack detection is still limited by many factors. First, existing automatic crack detection is unable to deal with challenges such as the varying texture of infrastructure surface, the complex topology of crack structure, the diverse characteristics of crack features, and random background noises. Also, for some application scenarios, it is not easy to acquire the crack images as the fracture could occur inside the surface. As such, it is necessary to improve crack detection by a fusion of multimodal data, which is typically disparate. MultiModal Machine Learning (MMML) achieves the ability to process and understand multi-source modal information through a machine learning approach. It accepts a variety of different input methods (e.g., language, image, voice, video) and, in some cases, produces different output modalities. There exist several attempts to improve the detection performance by integrating additional data of Nondestructive Testing (NDT) and Non-Destructive Evaluation (NDE) sensors, e.g., optical fiber sensing network (Wu et al., 2019), waveguide sensor (Wang et al., 2019), accelerometer (Gómez et al., 2016; Xiang and Liang, 2012), ultrasonic sensors (Wolf et al., 2015; Merazi-Meksen et al., 2014) and Ground Penetrating Radar (GPR) (Li et al., 2021). Despite the above-mentioned efforts for crack detection that relies on multimodal data, research in this field is still in its early stages.

In addition to crack detection algorithms, some pre-processing (e.g., image stitching) and post-processing (e.g., vision calibration) are also crucial to the final results. For instance, the work in Wu et al. (2021) proposed an image stitching method based on Oriented FAST and Rotated BRIEF feature matching algorithm for civil structures crack detection to overcome the limitation of the field of view of the camera deployed in the application. A tunnel-lining crack recognition system is developed in Lei et al. (2020). They proposed a self-regulating calibration method that uses parallel projection to realize crack characterization, which achieved real-time size calibration. The method can improve the stability and reliability of the crack detection system.

On the other hand, although the types and locations of cracks are crucial indicators, they are insufficient to characterize the structure conditions for many scenarios as they only contain the raw description. It is also important to fully explore and understand the crack properties, i.e., width, length, type, location, and orientation, which are the key parameters to describe the fracture process and critically rate/assess the severity of the structures. Well-designed crack properties and features can actually help to reduce the data volume required by the inspection task.

6. Conclusion

Accurate and automatic detection of cracks is critical for monitoring and guaranteeing the safety conditions of civil infrastructures. As such, it has attracted significant attention in the research community, and remarkable achievements have been made in crack detection using computer vision techniques over the past 20 years. In this survey, we aim to help relevant researchers and institutions summarize the current studies for civil infrastructures and identify future research directions. This survey focused on the major components of the vision-based crack detection framework, which are widely used for civil infrastructures' condition inspection and assessment.

To support a better understanding of the existing designs and comparison among various crack detection algorithms, we provided a rigorous discussion on data acquisition techniques, publicly available datasets, and evaluation metrics of crack detection algorithms. We provide a new taxonomy based on the development of automatic crack detection algorithms, i.e., threshold-based, morphology-based, ML-based, and DL-based approaches. Methods in these four categories detect cracks relied on pixel-intensity features, crack structure features, hand-crafted features, and automatically learned features, respectively.

In summary, the threshold-based methods generally run fast but are sensitive to noises and illumination. The morphology-based methods have been widely used in earlier works for detecting the structures, sizes, and contours of the cracks. However, methods in this category usually cannot handle crack images with complex backgrounds, and some of them are very time-consuming. The machine learning techniques proved to be more efficient than the first two categories. Earlier works that employed ML techniques typically relied on hand-crafted features and regarded crack detection as a classification problem, which improved the detection accuracy but significantly decreased the running time efficiency. Also, their robustness depends on the quality of hand-crafted features, which further requires strong domain knowledge. Recently, most attention has been paid to DL-based methods, which achieved steady improvements in detection accuracy and speed while automatically learning required data features in an end-to-end manner.

Even though the DL-based methods have been successfully applied to many aspects of crack detection problems, such as crack classification, localization, and segmentation, several challenging issues remain resolved. We have reviewed the unsolved issues and main challenges, and provided some insights on future prospects.

CRediT authorship contribution statement

Dihao Ai: Search of related papers, Methodology, Writing – original draft. **Guiyuan Jiang:** Methodology, Search of related works, Writing – original draft, Paper revision. **Siew-Kei Lam:** Supervision, Writing – review & editing. **Peilan He:** Conceptualization, Visualization, Writing – original draft. **Chengwu Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This is a survey paper. The links to available public datasets have been discussed in paper.

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